Tab 1

HW3 Documentation

Results:

Image classification MNIST: 98.52%

```
Epoch 0, Iteration 0, loss = 2.2964
Epoch 0, Iteration 100, loss = 0.4279
Epoch 0, Iteration 200, loss = 0.2643
Epoch 0, Iteration 300, loss = 0.2244
Epoch 0, Iteration 400, loss = 0.1880
Eval 9415 / 10000 correct (94.15)
Epoch 1, Iteration 0, loss = 0.2398
Epoch 1, Iteration 100, loss = 0.2263
Epoch 1, Iteration 200, loss = 0.1790
Epoch 1, Iteration 300, loss = 0.1663
Epoch 1, Iteration 400, loss = 0.1186
Eval 9572 / 10000 correct (95.72)
Epoch 2, Iteration 0, loss = 0.1595
Epoch 2, Iteration 100, loss = 0.0701
Epoch 2, Iteration 200, loss = 0.1148
Epoch 2, Iteration 300, loss = 0.1091
Epoch 2, Iteration 400, loss = 0.1122
Eval 9695 / 10000 correct (96.95)
Epoch 3, Iteration 0, loss = 0.0636
Epoch 3, Iteration 100, loss = 0.0736
Epoch 3, Iteration 200, loss = 0.1786
Epoch 3, Iteration 300, loss = 0.1464
Epoch 3, Iteration 400, loss = 0.0662
Eval 9719 / 10000 correct (97.19)
Epoch 4, Iteration 0, loss = 0.0515
...
Epoch 99, Iteration 200, loss = 0.0005
Epoch 99, Iteration 300, loss = 0.0002
Epoch 99, Iteration 400, loss = 0.0002
Eval 9852 / 10000 correct (98.52)
```

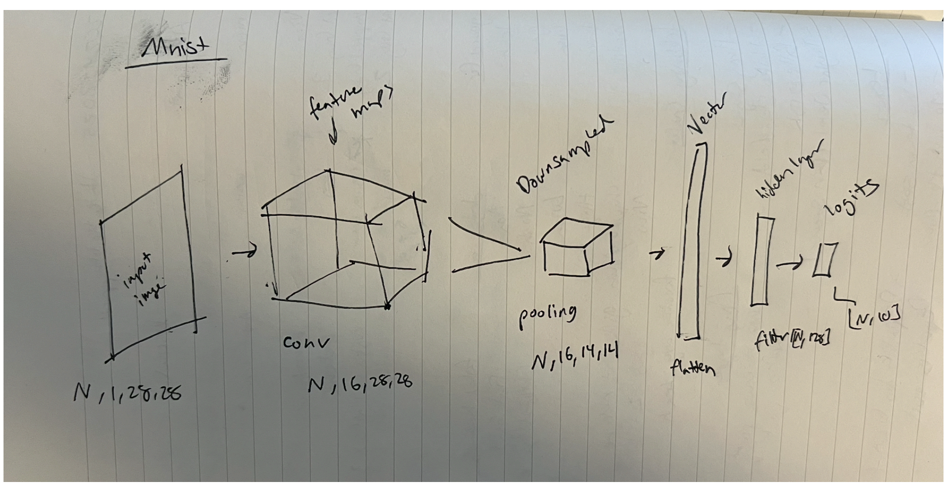
Image classification CIFAR: 76.34%

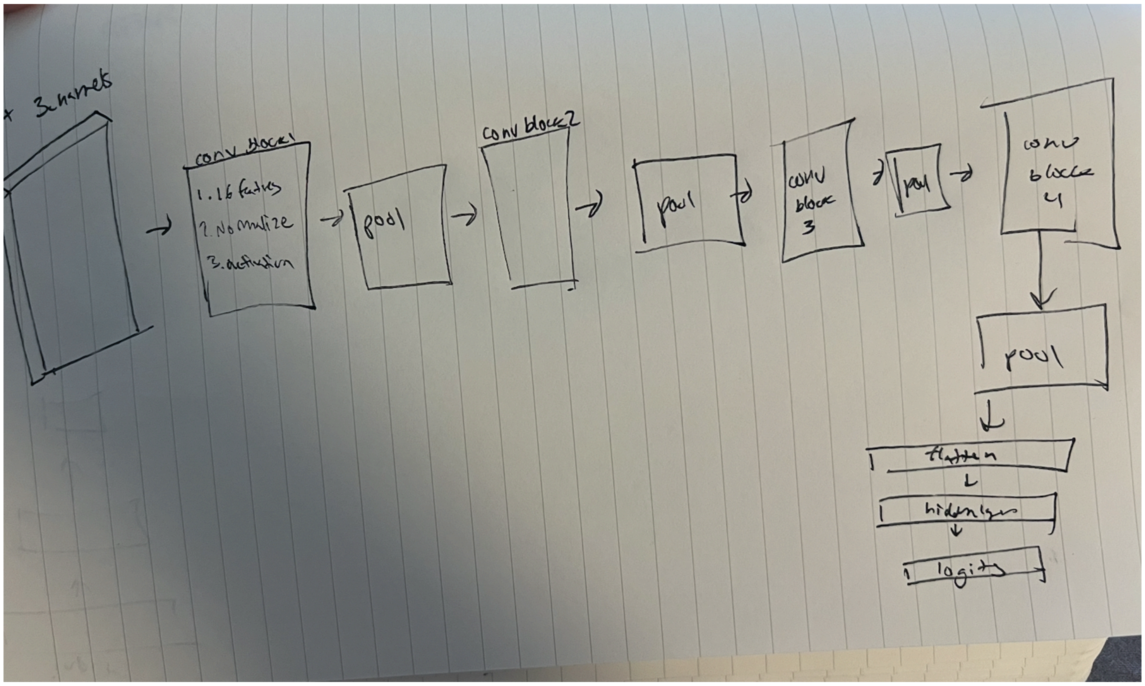
```
Epoch 94, Iteration 0, loss = 0.0001
Epoch 94, Iteration 100, loss = 0.0000
Epoch 94, Iteration 200, loss = 0.0000
Epoch 94, Iteration 300, loss = 0.0000
Eval 7625 / 10000 correct (76.25)
Epoch 95, Iteration 0, loss = 0.0001
Epoch 95, Iteration 100, loss = 0.0000
Epoch 95, Iteration 200, loss = 0.0001
Epoch 95, Iteration 300, loss = 0.0001
Eval 7630 / 10000 correct (76.30)
Epoch 96, Iteration 0, loss = 0.0001
Epoch 96, Iteration 100, loss = 0.0000
Epoch 96, Iteration 200, loss = 0.0000
Epoch 96, Iteration 300, loss = 0.0000
Eval 7629 / 10000 correct (76.29)
Epoch 97, Iteration 0, loss = 0.0001
Epoch 97, Iteration 100, loss = 0.0000
Epoch 97, Iteration 200, loss = 0.0000
Epoch 97, Iteration 300, loss = 0.0000
Eval 7634 / 10000 correct (76.34)
Epoch 98, Iteration 0, loss = 0.0000
Epoch 98, Iteration 100, loss = 0.0000
Epoch 98, Iteration 200, loss = 0.0000
Epoch 98, Iteration 300, loss = 0.0000
Eval 7636 / 10000 correct (76.36)
Epoch 99, Iteration 0, loss = 0.0000
Epoch 99, Iteration 100, loss = 0.0000
Epoch 99, Iteration 200, loss = 0.0000
Epoch 99, Iteration 300, loss = 0.0000
Eval 7634 / 10000 correct (76.34)
```

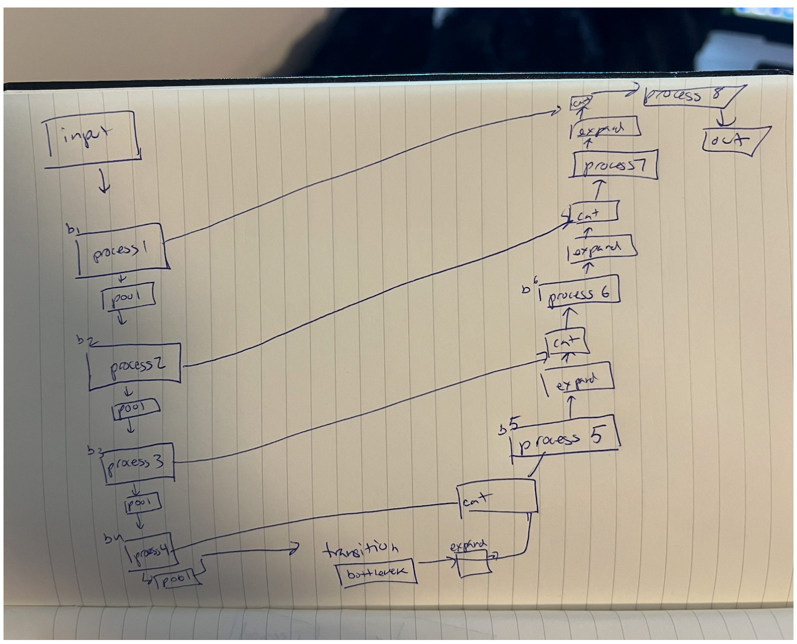
Image segmentation final results: 74.78%

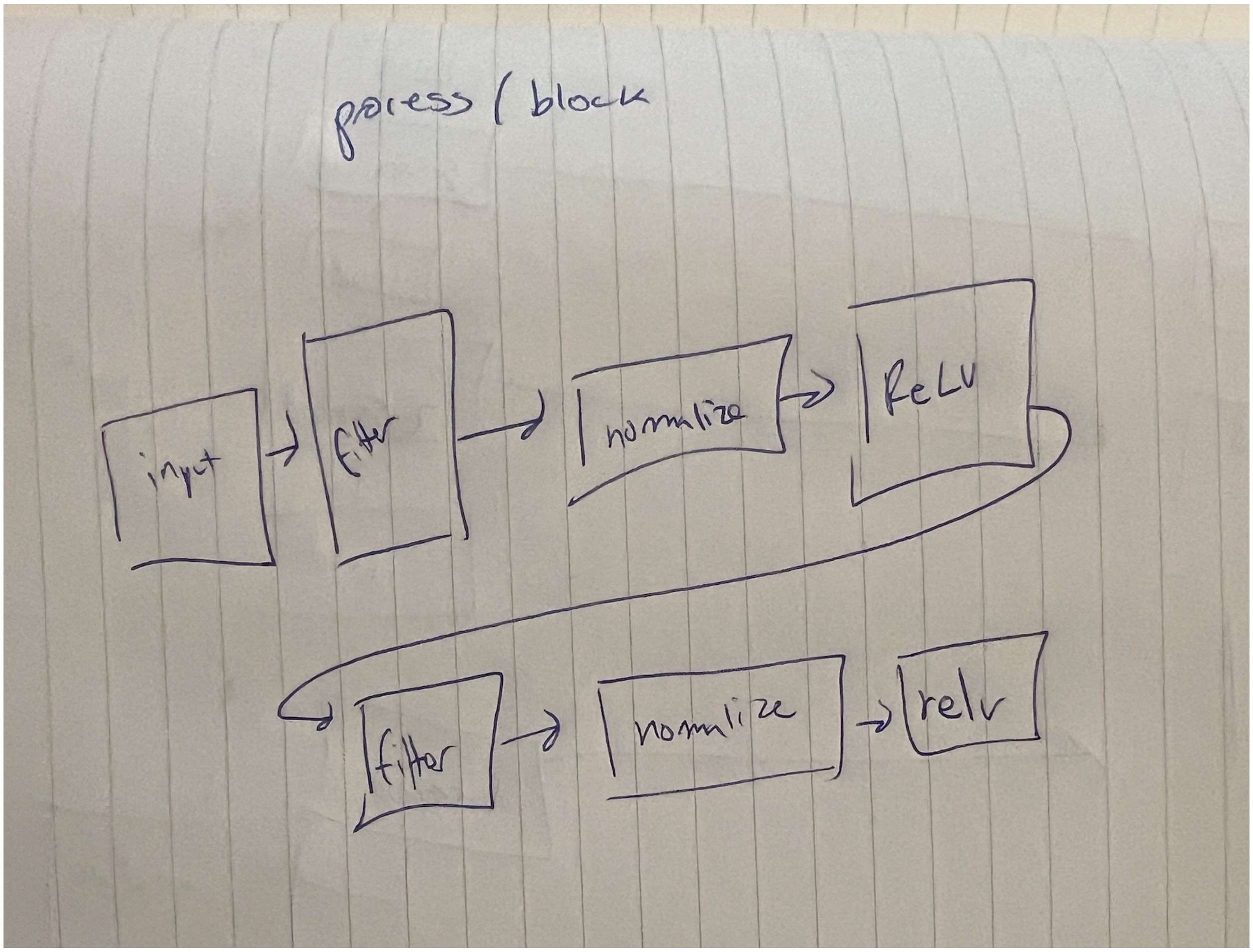
```
Epoch 95, Iteration 300, loss = 0.0141
Epoch 95, Iteration 400, loss = 0.0604
Epoch 95, Iteration 500, loss = 0.0217
Epoch 95, Iteration 600, loss = 0.0147
Epoch 95, Iteration 700, loss = 0.0560
Eval 7648755 / 10240000 correct (74.69)
Epoch 96, Iteration 0, loss = 0.0116
Epoch 96, Iteration 100, loss = 0.0135
Epoch 96, Iteration 200, loss = 0.0134
Epoch 96, Iteration 300, loss = 0.0165
Epoch 96, Iteration 400, loss = 0.0538
Epoch 96, Iteration 500, loss = 0.0506
Epoch 96, Iteration 600, loss = 0.0169
Epoch 96, Iteration 700, loss = 0.0100
Eval 7617463 / 10240000 correct (74.39)
Epoch 97, Iteration 0, loss = 0.0092
Epoch 97, Iteration 100, loss = 0.0315
Epoch 97, Iteration 200, loss = 0.0472
Epoch 97, Iteration 300, loss = 0.0173
Epoch 97, Iteration 400, loss = 0.0152
Epoch 97, Iteration 500, loss = 0.0196
Epoch 97, Iteration 600, loss = 0.0498
Epoch 97, Iteration 700, loss = 0.0125
Eval 7620842 / 10240000 correct (74.42)
Epoch 98, Iteration 0, loss = 0.0978
Epoch 98, Iteration 100, loss = 0.0109
Epoch 98, Iteration 200, loss = 0.0908
Epoch 98, Iteration 300, loss = 0.0088
Epoch 98, Iteration 400, loss = 0.0196
Epoch 98, Iteration 500, loss = 0.0164
Epoch 98, Iteration 600, loss = 0.0160
Epoch 98, Iteration 700, loss = 0.0783
Eval 7571793 / 10240000 correct (73.94)
Epoch 99, Iteration 0, loss = 0.0141
Epoch 99, Iteration 100, loss = 0.0170
Epoch 99, Iteration 200, loss = 0.0601
Epoch 99, Iteration 300, loss = 0.0227
Epoch 99, Iteration 400, loss = 0.0663
Epoch 99, Iteration 500, loss = 0.0465
Epoch 99, Iteration 600, loss = 0.0341
Epoch 99, Iteration 700, loss = 0.0879
Eval 7657923 / 10240000 correct (74.78)
```

Diagrams:




3 channels
conv block1
1. 16 features
2. Normalize
3. activation
pool
conv block2
pool
conv block 3
pool
conv block 4
pool
flatten
hidden layers
logits


input
process 1
pool
process 2
pool
process 3
pool
process 4
pool
transition
bottleneck
expand
cat
process 5
expand
cat
process 6
expand
cat
process 7
expand
cat
process 8
out



Design choices:

<u>Data Preparation:</u> Increased batch size from 4 to 128 to make use of GPU space

<u>MNIST Approach</u>:

First started with two filters:

```
class myNet(nn.Module):
   def __init__(self):
       super(myNet, self).__init__()
       # Set up your own CNN.
       num_classes = 20
       self.conv1 = nn.Conv2d(3, 32, kernel_size = 3, padding = 1)
       self.conv2 = nn.Conv2d(32, num_classes, kernel_size = 1)
```

```python
    def forward(self, x):
        # forward
        x = F.relu(self.conv1(x))
        out = self.conv2(x)
        return out
```

But this plateaued at a 19.65 accuracy
Used Relu activation as class slides mention it to be the best choice.

Filters stayed in the 3x3 size and used powers of 2 when deciding feature count for filtering

Second MNIST:
Keeping Convolution for feature extraction, adding pooling for spatial downsampling and fully connected layers to map features to the class logits.

```python
class MNIST_models(myNet):
    def __init__(self):
        super(MNIST_models, self).__init__()
        self.num_classes = 10
        self.conv1 = nn.Conv2d(1, 16, 3, padding=1)
        self.pool = nn.MaxPool2d(2, 2)
        self.fc1 = nn.Linear(16 * 14 * 14, 128)
        self.fc2 = nn.Linear(128, self.num_classes)

        # Set up your own CNN for MNIST.

    def forward(self, x):
        # forward
        x = F.relu(self.conv1(x))
        x = self.pool(x)
        x = x.view(x.size(0), -1)
        x = F.relu(self.fc1(x))
        out = self.fc2(x)
        return out
```

```
model = MNIST_models()
optimizer = optim.SGD(model.parameters(), lr, momentum, weight_decay)
train(mnist_loader_train, mnist_loader_test, model, optimizer, epochs=100)
```
[9] 26.0s

```
Epoch 0, Iteration 0, loss = 2.3043
Epoch 0, Iteration 100, loss = 0.4201
Epoch 0, Iteration 200, loss = 0.4613
Epoch 0, Iteration 300, loss = 0.2811
Epoch 0, Iteration 400, loss = 0.1680
Eval 9363 / 10000 correct (93.63)
Epoch 1, Iteration 0, loss = 0.3048
Epoch 1, Iteration 100, loss = 0.1341
Epoch 1, Iteration 200, loss = 0.1357
Epoch 1, Iteration 300, loss = 0.0959
Epoch 1, Iteration 400, loss = 0.3320
Eval 9624 / 10000 correct (96.24)
Epoch 2, Iteration 0, loss = 0.0977
Epoch 2, Iteration 100, loss = 0.1451
Epoch 2, Iteration 200, loss = 0.1515
Epoch 2, Iteration 300, loss = 0.1597
Epoch 2, Iteration 400, loss = 0.0891
Eval 9657 / 10000 correct (96.57)
Epoch 3, Iteration 0, loss = 0.1259
Epoch 3, Iteration 100, loss = 0.0468
Epoch 3, Iteration 200, loss = 0.0287
Epoch 3, Iteration 300, loss = 0.1062
```

Immediately jumped to 96.57 success

In terms of Loss function, using **nn.CrossEntropyLoss()**

When trying other loss functions, improvement was much slower on the MNIST model.
Example below

```
train(cifar10_loader_train, cifar10_loader_test, model, optimiz
```
[15] 2m 9.2s

```
Epoch 0, Iteration 0, loss = 2.3032
Epoch 0, Iteration 100, loss = 1.9365
Epoch 0, Iteration 200, loss = 1.8515
Epoch 0, Iteration 300, loss = 1.5711
Eval 4529 / 10000 correct (45.29)
Epoch 1, Iteration 0, loss = 1.5099
Epoch 1, Iteration 100, loss = 1.6794
Epoch 1, Iteration 200, loss = 1.5587
Epoch 1, Iteration 300, loss = 1.4418
Eval 5264 / 10000 correct (52.64)
Epoch 2, Iteration 0, loss = 1.3263
Epoch 2, Iteration 100, loss = 1.2392
Epoch 2, Iteration 200, loss = 1.2701
Epoch 2, Iteration 300, loss = 1.2746
Eval 5401 / 10000 correct (54.01)
Epoch 3, Iteration 0, loss = 1.1774
Epoch 3, Iteration 100, loss = 1.2916
Epoch 3, Iteration 200, loss = 1.1135
Epoch 3, Iteration 300, loss = 1.2093
Eval 5827 / 10000 correct (58.27)
Epoch 4, Iteration 0, loss = 1.0287
Epoch 4, Iteration 100, loss = 1.1602
Epoch 4, Iteration 200, loss = 1.2079
Epoch 4, Iteration 300, loss = 1.0848
Eval 5925 / 10000 correct (59.25)
...
Epoch 15, Iteration 200, loss = 0.6642
Epoch 15, Iteration 300, loss = 0.5749
Eval 6310 / 10000 correct (63.10)
Epoch 16, Iteration 0, loss = 0.4402
```
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell o

<u>CIFAR Model:</u>

First, I tried the same model as the MNIST. This resulted in a jump to 60% correct but then improvement plateaued.

I then tried increasing the feature filter extraction from 16 to 32.

```
    train(cifar10_loader_train, cifar10_loader_test, model, o
22]   2m 16.8s

Epoch 6, Iteration 200, loss = 1.0893
Epoch 6, Iteration 300, loss = 0.8856
Eval 6345 / 10000 correct (63.45)
Epoch 7, Iteration 0, loss = 0.9246
Epoch 7, Iteration 100, loss = 0.8424
Epoch 7, Iteration 200, loss = 0.7200
Epoch 7, Iteration 300, loss = 0.8753
Eval 6475 / 10000 correct (64.75)
Epoch 8, Iteration 0, loss = 0.7334
Epoch 8, Iteration 100, loss = 0.8991
Epoch 8, Iteration 200, loss = 0.6671
Epoch 8, Iteration 300, loss = 0.6509
Eval 6547 / 10000 correct (65.47)
Epoch 9, Iteration 0, loss = 0.6835
Epoch 9, Iteration 100, loss = 0.7170
Epoch 9, Iteration 200, loss = 0.7489
Epoch 9, Iteration 300, loss = 0.7220
Eval 6582 / 10000 correct (65.82)
Epoch 10, Iteration 0, loss = 0.7675
Epoch 10, Iteration 100, loss = 0.7558
Epoch 10, Iteration 200, loss = 0.8323
Epoch 10, Iteration 300, loss = 0.7070
Eval 6397 / 10000 correct (63.97)
Epoch 11, Iteration 0, loss = 0.5802
Epoch 11, Iteration 100, loss = 0.6297
Epoch 11, Iteration 200, loss = 0.6500
Epoch 11, Iteration 300, loss = 0.6718
Eval 6530 / 10000 correct (65.30)
Epoch 12, Iteration 0, loss = 0.5392
Epoch 12, Iteration 100, loss = 0.5808
```

Oscillating around 65

Then I added another conv filters which resulted in osculating around the 68%:

```
optimizer = optim.SGD(model.parameters(), lr,
train(cifar10_loader_train, cifar10_loader_tes
3m 10.1s
Epoch 7, Iteration 100, loss = 0.7793
Epoch 7, Iteration 200, loss = 0.7459
Epoch 7, Iteration 300, loss = 0.9797
Eval 6606 / 10000 correct (66.06)
Epoch 8, Iteration 0, loss = 0.7313
Epoch 8, Iteration 100, loss = 0.8522
Epoch 8, Iteration 200, loss = 0.7432
Epoch 8, Iteration 300, loss = 0.8606
Eval 6760 / 10000 correct (67.60)
Epoch 9, Iteration 0, loss = 0.7889
Epoch 9, Iteration 100, loss = 0.6726
Epoch 9, Iteration 200, loss = 0.8406
Epoch 9, Iteration 300, loss = 0.7017
Eval 6750 / 10000 correct (67.50)
Epoch 10, Iteration 0, loss = 0.6610
Epoch 10, Iteration 100, loss = 0.7975
Epoch 10, Iteration 200, loss = 0.6690
Epoch 10, Iteration 300, loss = 0.7211
Eval 6837 / 10000 correct (68.37)
Epoch 11, Iteration 0, loss = 0.5985
Epoch 11, Iteration 100, loss = 0.6576
Epoch 11, Iteration 200, loss = 0.6486
Epoch 11, Iteration 300, loss = 0.7665
Eval 6706 / 10000 correct (67.06)
Epoch 12, Iteration 0, loss = 0.5628
Epoch 12, Iteration 100, loss = 0.5010
Epoch 12, Iteration 200, loss = 0.7514
Epoch 12, Iteration 300, loss = 0.6807
Eval 6798 / 10000 correct (67.98)
Epoch 13, Iteration 0, loss = 0.4930
```

Next I added another filters and included pooling between the filters

```
model = CIFAR10_models()
optimizer = optim.SGD(model.parameters(), lr, momentum,
train(cifar10_loader_train, cifar10_loader_test, model,
[35] 4m 20.2s
Epoch 7, Iteration 100, loss = 0.8719
Epoch 7, Iteration 200, loss = 0.8407
Epoch 7, Iteration 300, loss = 0.8088
Eval 6741 / 10000 correct (67.41)
Epoch 8, Iteration 0, loss = 0.8529
Epoch 8, Iteration 100, loss = 0.9395
Epoch 8, Iteration 200, loss = 0.8154
Epoch 8, Iteration 300, loss = 0.9537
Eval 6715 / 10000 correct (67.15)
Epoch 9, Iteration 0, loss = 0.7391
Epoch 9, Iteration 100, loss = 0.8628
Epoch 9, Iteration 200, loss = 0.7512
Epoch 9, Iteration 300, loss = 0.9789
Eval 6904 / 10000 correct (69.04)
Epoch 10, Iteration 0, loss = 0.7452
Epoch 10, Iteration 100, loss = 0.8784
Epoch 10, Iteration 200, loss = 0.7744
Epoch 10, Iteration 300, loss = 0.7687
Eval 6924 / 10000 correct (69.24)
Epoch 11, Iteration 0, loss = 0.6709
Epoch 11, Iteration 100, loss = 0.6446
Epoch 11, Iteration 200, loss = 0.5719
Epoch 11, Iteration 300, loss = 0.5779
Eval 7031 / 10000 correct (70.31)
Epoch 12, Iteration 0, loss = 0.6126
Epoch 12, Iteration 100, loss = 0.6708
Epoch 12, Iteration 200, loss = 0.5677
Epoch 12, Iteration 300, loss = 0.5490
Eval 6980 / 10000 correct (69.80)
Epoch 13, Iteration 0, loss = 0.5380
```

Now osculating around 70

Then I added more filters and normalization between the filters. This resulted in worse performance:

```
1m 23.4s

Epoch 0, Iteration 0, loss = 2.3020
Epoch 0, Iteration 100, loss = 1.6003
Epoch 0, Iteration 200, loss = 1.4729
Epoch 0, Iteration 300, loss = 1.3085
Eval 5345 / 10000 correct (53.45)
Epoch 1, Iteration 0, loss = 1.0583
Epoch 1, Iteration 100, loss = 1.2819
Epoch 1, Iteration 200, loss = 1.0067
Epoch 1, Iteration 300, loss = 0.9468
Eval 6606 / 10000 correct (66.06)
Epoch 2, Iteration 0, loss = 1.0063
Epoch 2, Iteration 100, loss = 0.7442
Epoch 2, Iteration 200, loss = 0.9043
Epoch 2, Iteration 300, loss = 0.9050
Eval 5837 / 10000 correct (58.37)
Epoch 3, Iteration 0, loss = 0.6223
Epoch 3, Iteration 100, loss = 0.6941
```

Removed some filters and repositioned pooling to end of filters and normalization:

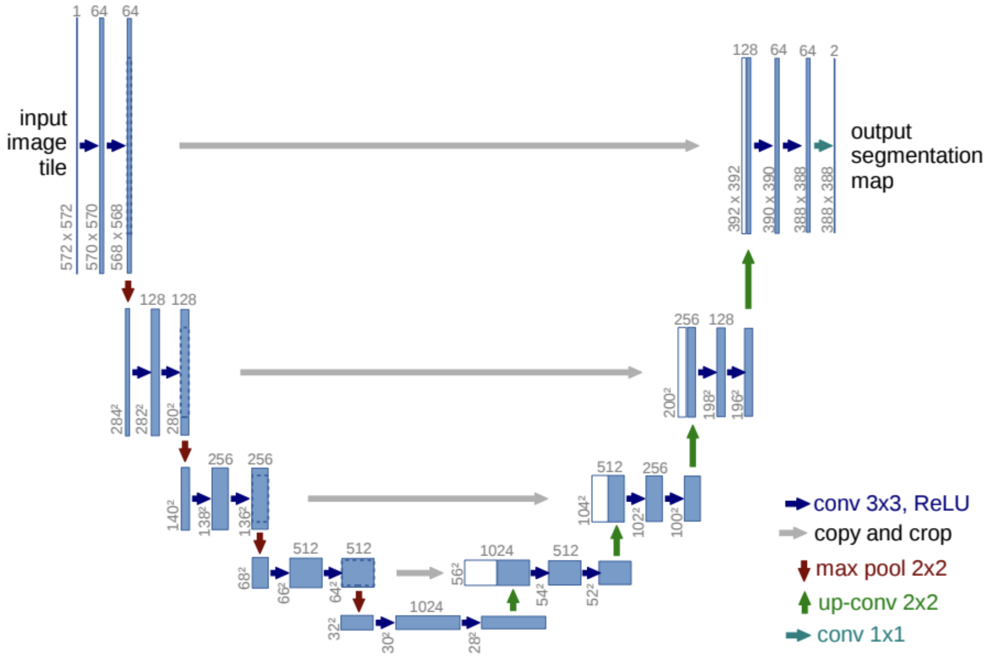
```python
class CIFAR10_models(myNet):
   def __init__(self):
       super(CIFAR10_models, self).__init__()
       self.num_classes = 10
       self.conv1 = nn.Conv2d(3, 16, 3, padding=1)
       self.bn1 = nn.BatchNorm2d(16)
       self.conv2 = nn.Conv2d(16, 32, 3, padding=1)
       self.bn2 = nn.BatchNorm2d(32)
       self.conv3 = nn.Conv2d(32, 64, 3, padding=1)
       self.bn3 = nn.BatchNorm2d(64)
       self.conv4 = nn.Conv2d(64, 128, 3, padding=1)
       self.pool = nn.MaxPool2d(2, 2)
       self.fc1 = nn.Linear(128 * 2 * 2, 128)
       self.fc2 = nn.Linear(128, self.num_classes)
       self.model_save_path = "./model"
```

```
optimizer = optim.SGD(model.parameters(), lr, moment
train(cifar10_loader_train, cifar10_loader_test, mod
[48]   3m 40.1s
...
Eval 5837 / 10000 correct (58.37)
Epoch 3, Iteration 0, loss = 0.6223
Epoch 3, Iteration 100, loss = 0.6941
Epoch 3, Iteration 200, loss = 0.6890
Epoch 3, Iteration 300, loss = 0.7166
Eval 6651 / 10000 correct (66.51)
Epoch 4, Iteration 0, loss = 0.6596
Epoch 4, Iteration 100, loss = 0.6736
Epoch 4, Iteration 200, loss = 0.6634
Epoch 4, Iteration 300, loss = 0.6386
Eval 7133 / 10000 correct (71.33)
Epoch 5, Iteration 0, loss = 0.5199
Epoch 5, Iteration 100, loss = 0.5940
Epoch 5, Iteration 200, loss = 0.5632
Epoch 5, Iteration 300, loss = 0.6260
Eval 7002 / 10000 correct (70.02)
Epoch 6, Iteration 0, loss = 0.3773
Epoch 6, Iteration 100, loss = 0.4922
Epoch 6, Iteration 200, loss = 0.6911
Epoch 6, Iteration 300, loss = 0.6923
Eval 7466 / 10000 correct (74.66)
Epoch 7, Iteration 0, loss = 0.4163
Epoch 7, Iteration 100, loss = 0.3027
Epoch 7, Iteration 200, loss = 0.4931
Epoch 7, Iteration 300, loss = 0.6091
Eval 7393 / 10000 correct (73.93)
Epoch 8, Iteration 0, loss = 0.5386
Epoch 8, Iteration 100, loss = 0.5145
Epoch 8, Iteration 200, loss = 0.3596
Epoch 8, Iteration 300, loss = 0.4075
```

<u>Image Segmentation</u>

At first I tried to use a similar CNN architect as image classification. This proved not to work and plateaued around 50%s at best. I also tried different step sizes to see if over shooting was resulting in poor accuracy. Other designs I explored were using a different optimizer -> The Adam optimizer (listed in slides). I switched to using Google Colab in order to run on a GPU and speed training. I ended up starting from scratch and looking at the U-Net structure and implementing a version of this.

Using a similar strategy as above, I started small with a few layers and worked my way up to more and more layers.



From UNet: convolution Networks for Biomedical Image Segmentation paper linked

Other resources as reference used while working on the project

https://www.youtube.com/watch?v=IHq1t7NxS8k

https://www.youtube.com/watch?v=CtzfbUwrYGI&t=1175s

Tab 2

CIFAR training output

```
 Epoch 0, Iteration 0, loss = 2.3115
Epoch 0, Iteration 100, loss = 1.5465
Epoch 0, Iteration 200, loss = 1.5448
Epoch 0, Iteration 300, loss = 1.1991
Eval 5598 / 10000 correct (55.98)
Epoch 1, Iteration 0, loss = 0.8527
Epoch 1, Iteration 100, loss = 0.9723
Epoch 1, Iteration 200, loss = 0.9968
Epoch 1, Iteration 300, loss = 0.8683
Eval 5852 / 10000 correct (58.52)
Epoch 2, Iteration 0, loss = 0.8200
Epoch 2, Iteration 100, loss = 0.8857
Epoch 2, Iteration 200, loss = 1.0516
Epoch 2, Iteration 300, loss = 0.8786
Eval 5874 / 10000 correct (58.74)
Epoch 3, Iteration 0, loss = 0.6816
Epoch 3, Iteration 100, loss = 0.7650
Epoch 3, Iteration 200, loss = 0.5387
Epoch 3, Iteration 300, loss = 0.7702
Eval 6550 / 10000 correct (65.50)
Epoch 4, Iteration 0, loss = 0.7999
Epoch 4, Iteration 100, loss = 0.5460
Epoch 4, Iteration 200, loss = 0.6170
Epoch 4, Iteration 300, loss = 0.7412
Eval 7016 / 10000 correct (70.16)
Epoch 5, Iteration 0, loss = 0.6495
Epoch 5, Iteration 100, loss = 0.5981
Epoch 5, Iteration 200, loss = 0.5316
Epoch 5, Iteration 300, loss = 0.7187
Eval 7017 / 10000 correct (70.17)
Epoch 6, Iteration 0, loss = 0.5376
Epoch 6, Iteration 100, loss = 0.5585
Epoch 6, Iteration 200, loss = 0.5445
Epoch 6, Iteration 300, loss = 0.5623
Eval 7118 / 10000 correct (71.18)
Epoch 7, Iteration 0, loss = 0.4608
Epoch 7, Iteration 100, loss = 0.4733
Epoch 7, Iteration 200, loss = 0.5064
Epoch 7, Iteration 300, loss = 0.3627
Eval 7431 / 10000 correct (74.31)
Epoch 8, Iteration 0, loss = 0.4302
Epoch 8, Iteration 100, loss = 0.4105
Epoch 8, Iteration 200, loss = 0.4535
Epoch 8, Iteration 300, loss = 0.4189
Eval 7172 / 10000 correct (71.72)
Epoch 9, Iteration 0, loss = 0.3618
Epoch 9, Iteration 100, loss = 0.3603
Epoch 9, Iteration 200, loss = 0.4137
Epoch 9, Iteration 300, loss = 0.5148
Eval 7333 / 10000 correct (73.33)
Epoch 10, Iteration 0, loss = 0.2651
```

```
Epoch 10, Iteration 100, loss = 0.4874
Epoch 10, Iteration 200, loss = 0.3907
Epoch 10, Iteration 300, loss = 0.3872
Eval 7439 / 10000 correct (74.39)
Epoch 11, Iteration 0, loss = 0.2916
Epoch 11, Iteration 100, loss = 0.2909
Epoch 11, Iteration 200, loss = 0.4688
Epoch 11, Iteration 300, loss = 0.3506
Eval 7402 / 10000 correct (74.02)
Epoch 12, Iteration 0, loss = 0.2917
Epoch 12, Iteration 100, loss = 0.2113
Epoch 12, Iteration 200, loss = 0.4386
Epoch 12, Iteration 300, loss = 0.2828
Eval 7028 / 10000 correct (70.28)
Epoch 13, Iteration 0, loss = 0.3369
Epoch 13, Iteration 100, loss = 0.2997
Epoch 13, Iteration 200, loss = 0.3169
Epoch 13, Iteration 300, loss = 0.3464
Eval 7279 / 10000 correct (72.79)
Epoch 14, Iteration 0, loss = 0.2578
Epoch 14, Iteration 100, loss = 0.1858
Epoch 14, Iteration 200, loss = 0.2388
Epoch 14, Iteration 300, loss = 0.2232
Eval 7185 / 10000 correct (71.85)
Epoch 15, Iteration 0, loss = 0.1777
Epoch 15, Iteration 100, loss = 0.1826
Epoch 15, Iteration 200, loss = 0.2601
Epoch 15, Iteration 300, loss = 0.2827
Eval 7305 / 10000 correct (73.05)
Epoch 16, Iteration 0, loss = 0.1396
Epoch 16, Iteration 100, loss = 0.1146
Epoch 16, Iteration 200, loss = 0.1411
Epoch 16, Iteration 300, loss = 0.2899
Eval 6638 / 10000 correct (66.38)
Epoch 17, Iteration 0, loss = 0.1283
Epoch 17, Iteration 100, loss = 0.1472
Epoch 17, Iteration 200, loss = 0.1530
Epoch 17, Iteration 300, loss = 0.2378
Eval 7184 / 10000 correct (71.84)
Epoch 18, Iteration 0, loss = 0.2156
Epoch 18, Iteration 100, loss = 0.0945
Epoch 18, Iteration 200, loss = 0.0922
Epoch 18, Iteration 300, loss = 0.2791
Eval 7220 / 10000 correct (72.20)
Epoch 19, Iteration 0, loss = 0.1068
Epoch 19, Iteration 100, loss = 0.1195
Epoch 19, Iteration 200, loss = 0.0751
Epoch 19, Iteration 300, loss = 0.0817
Eval 7411 / 10000 correct (74.11)
Epoch 20, Iteration 0, loss = 0.0943
Epoch 20, Iteration 100, loss = 0.1893
Epoch 20, Iteration 200, loss = 0.0702
Epoch 20, Iteration 300, loss = 0.1327
Eval 7428 / 10000 correct (74.28)
Epoch 21, Iteration 0, loss = 0.1310
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Epoch 21, Iteration 100, loss = 0.0935
Epoch 21, Iteration 200, loss = 0.0853
Epoch 21, Iteration 300, loss = 0.1040
Eval 7179 / 10000 correct (71.79)
Epoch 22, Iteration 0, loss = 0.0588
Epoch 22, Iteration 100, loss = 0.0416
Epoch 22, Iteration 200, loss = 0.1335
Epoch 22, Iteration 300, loss = 0.1140
Eval 7339 / 10000 correct (73.39)
Epoch 23, Iteration 0, loss = 0.1095
Epoch 23, Iteration 100, loss = 0.0729
Epoch 23, Iteration 200, loss = 0.0507
Epoch 23, Iteration 300, loss = 0.1075
Eval 7364 / 10000 correct (73.64)
Epoch 24, Iteration 0, loss = 0.0719
Epoch 24, Iteration 100, loss = 0.0257
Epoch 24, Iteration 200, loss = 0.0991
Epoch 24, Iteration 300, loss = 0.1236
Eval 7286 / 10000 correct (72.86)
Epoch 25, Iteration 0, loss = 0.0881
Epoch 25, Iteration 100, loss = 0.0518
Epoch 25, Iteration 200, loss = 0.0772
Epoch 25, Iteration 300, loss = 0.0908
Eval 7303 / 10000 correct (73.03)
Epoch 26, Iteration 0, loss = 0.1126
Epoch 26, Iteration 100, loss = 0.0546
Epoch 26, Iteration 200, loss = 0.1447
Epoch 26, Iteration 300, loss = 0.0638
Eval 7350 / 10000 correct (73.50)
Epoch 27, Iteration 0, loss = 0.0948
Epoch 27, Iteration 100, loss = 0.0254
Epoch 27, Iteration 200, loss = 0.0584
Epoch 27, Iteration 300, loss = 0.0842
Eval 7444 / 10000 correct (74.44)
Epoch 28, Iteration 0, loss = 0.0338
Epoch 28, Iteration 100, loss = 0.0495
Epoch 28, Iteration 200, loss = 0.0450
Epoch 28, Iteration 300, loss = 0.1162
Eval 7324 / 10000 correct (73.24)
Epoch 29, Iteration 0, loss = 0.0355
Epoch 29, Iteration 100, loss = 0.0474
Epoch 29, Iteration 200, loss = 0.0363
Epoch 29, Iteration 300, loss = 0.0345
Eval 7423 / 10000 correct (74.23)
Epoch 30, Iteration 0, loss = 0.0267
Epoch 30, Iteration 100, loss = 0.0374
Epoch 30, Iteration 200, loss = 0.0472
Epoch 30, Iteration 300, loss = 0.0296
Eval 7267 / 10000 correct (72.67)
Epoch 31, Iteration 0, loss = 0.0840
Epoch 31, Iteration 100, loss = 0.0336
Epoch 31, Iteration 200, loss = 0.0304
Epoch 31, Iteration 300, loss = 0.0296
Eval 7364 / 10000 correct (73.64)
Epoch 32, Iteration 0, loss = 0.0489
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Epoch 32, Iteration 100, loss = 0.0364
Epoch 32, Iteration 200, loss = 0.0212
Epoch 32, Iteration 300, loss = 0.0631
Eval 7514 / 10000 correct (75.14)
Epoch 33, Iteration 0, loss = 0.0264
Epoch 33, Iteration 100, loss = 0.0366
Epoch 33, Iteration 200, loss = 0.0195
Epoch 33, Iteration 300, loss = 0.0783
Eval 7209 / 10000 correct (72.09)
Epoch 34, Iteration 0, loss = 0.1324
Epoch 34, Iteration 100, loss = 0.0155
Epoch 34, Iteration 200, loss = 0.0189
Epoch 34, Iteration 300, loss = 0.1322
Eval 7443 / 10000 correct (74.43)
Epoch 35, Iteration 0, loss = 0.0072
Epoch 35, Iteration 100, loss = 0.0144
Epoch 35, Iteration 200, loss = 0.0584
Epoch 35, Iteration 300, loss = 0.0383
Eval 7449 / 10000 correct (74.49)
Epoch 36, Iteration 0, loss = 0.0449
Epoch 36, Iteration 100, loss = 0.0296
Epoch 36, Iteration 200, loss = 0.0261
Epoch 36, Iteration 300, loss = 0.0280
Eval 7484 / 10000 correct (74.84)
Epoch 37, Iteration 0, loss = 0.0281
Epoch 37, Iteration 100, loss = 0.0502
Epoch 37, Iteration 200, loss = 0.0331
Epoch 37, Iteration 300, loss = 0.0432
Eval 7441 / 10000 correct (74.41)
Epoch 38, Iteration 0, loss = 0.0280
Epoch 38, Iteration 100, loss = 0.0590
Epoch 38, Iteration 200, loss = 0.0288
Epoch 38, Iteration 300, loss = 0.0591
Eval 7445 / 10000 correct (74.45)
Epoch 39, Iteration 0, loss = 0.0067
Epoch 39, Iteration 100, loss = 0.0091
Epoch 39, Iteration 200, loss = 0.0230
Epoch 39, Iteration 300, loss = 0.0156
Eval 7454 / 10000 correct (74.54)
Epoch 40, Iteration 0, loss = 0.0254
Epoch 40, Iteration 100, loss = 0.0044
Epoch 40, Iteration 200, loss = 0.0318
Epoch 40, Iteration 300, loss = 0.0305
Eval 7436 / 10000 correct (74.36)
Epoch 41, Iteration 0, loss = 0.0128
Epoch 41, Iteration 100, loss = 0.0187
Epoch 41, Iteration 200, loss = 0.0132
Epoch 41, Iteration 300, loss = 0.0036
Eval 7454 / 10000 correct (74.54)
Epoch 42, Iteration 0, loss = 0.0303
Epoch 42, Iteration 100, loss = 0.0107
Epoch 42, Iteration 200, loss = 0.0775
Epoch 42, Iteration 300, loss = 0.0379
Eval 7476 / 10000 correct (74.76)
Epoch 43, Iteration 0, loss = 0.0031
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Epoch 43, Iteration 100, loss = 0.0015
Epoch 43, Iteration 200, loss = 0.0186
Epoch 43, Iteration 300, loss = 0.0183
Eval 7470 / 10000 correct (74.70)
Epoch 44, Iteration 0, loss = 0.0069
Epoch 44, Iteration 100, loss = 0.0281
Epoch 44, Iteration 200, loss = 0.0137
Epoch 44, Iteration 300, loss = 0.0057
Eval 7470 / 10000 correct (74.70)
Epoch 45, Iteration 0, loss = 0.0289
Epoch 45, Iteration 100, loss = 0.0189
Epoch 45, Iteration 200, loss = 0.0436
Epoch 45, Iteration 300, loss = 0.0034
Eval 7502 / 10000 correct (75.02)
Epoch 46, Iteration 0, loss = 0.0650
Epoch 46, Iteration 100, loss = 0.0103
Epoch 46, Iteration 200, loss = 0.0158
Epoch 46, Iteration 300, loss = 0.0053
Eval 7421 / 10000 correct (74.21)
Epoch 47, Iteration 0, loss = 0.0903
Epoch 47, Iteration 100, loss = 0.0020
Epoch 47, Iteration 200, loss = 0.0098
Epoch 47, Iteration 300, loss = 0.0179
Eval 7474 / 10000 correct (74.74)
Epoch 48, Iteration 0, loss = 0.0112
Epoch 48, Iteration 100, loss = 0.0261
Epoch 48, Iteration 200, loss = 0.0124
Epoch 48, Iteration 300, loss = 0.0142
Eval 7470 / 10000 correct (74.70)
Epoch 49, Iteration 0, loss = 0.0509
Epoch 49, Iteration 100, loss = 0.0137
Epoch 49, Iteration 200, loss = 0.0412
Epoch 49, Iteration 300, loss = 0.0492
Eval 7482 / 10000 correct (74.82)
Epoch 50, Iteration 0, loss = 0.0223
Epoch 50, Iteration 100, loss = 0.0230
Epoch 50, Iteration 200, loss = 0.0297
Epoch 50, Iteration 300, loss = 0.0055
Eval 7451 / 10000 correct (74.51)
Epoch 51, Iteration 0, loss = 0.0350
Epoch 51, Iteration 100, loss = 0.0066
Epoch 51, Iteration 200, loss = 0.0212
Epoch 51, Iteration 300, loss = 0.0063
Eval 7504 / 10000 correct (75.04)
Epoch 52, Iteration 0, loss = 0.0061
Epoch 52, Iteration 100, loss = 0.0263
Epoch 52, Iteration 200, loss = 0.0175
Epoch 52, Iteration 300, loss = 0.0102
Eval 7559 / 10000 correct (75.59)
Epoch 53, Iteration 0, loss = 0.0031
Epoch 53, Iteration 100, loss = 0.0020
Epoch 53, Iteration 200, loss = 0.0089
Epoch 53, Iteration 300, loss = 0.0227
Eval 7581 / 10000 correct (75.81)
Epoch 54, Iteration 0, loss = 0.0031
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Epoch 54, Iteration 100, loss = 0.0034
Epoch 54, Iteration 200, loss = 0.0261
Epoch 54, Iteration 300, loss = 0.0296
Eval 7554 / 10000 correct (75.54)
Epoch 55, Iteration 0, loss = 0.0043
Epoch 55, Iteration 100, loss = 0.0022
Epoch 55, Iteration 200, loss = 0.0012
Epoch 55, Iteration 300, loss = 0.0006
Eval 7588 / 10000 correct (75.88)
Epoch 56, Iteration 0, loss = 0.0024
Epoch 56, Iteration 100, loss = 0.0079
Epoch 56, Iteration 200, loss = 0.0023
Epoch 56, Iteration 300, loss = 0.0007
Eval 7499 / 10000 correct (74.99)
Epoch 57, Iteration 0, loss = 0.0004
Epoch 57, Iteration 100, loss = 0.0010
Epoch 57, Iteration 200, loss = 0.0014
Epoch 57, Iteration 300, loss = 0.0042
Eval 7545 / 10000 correct (75.45)
Epoch 58, Iteration 0, loss = 0.0014
Epoch 58, Iteration 100, loss = 0.0006
Epoch 58, Iteration 200, loss = 0.0005
Epoch 58, Iteration 300, loss = 0.0012
Eval 7593 / 10000 correct (75.93)
Epoch 59, Iteration 0, loss = 0.0003
Epoch 59, Iteration 100, loss = 0.0001
Epoch 59, Iteration 200, loss = 0.0003
Epoch 59, Iteration 300, loss = 0.0005
Eval 7607 / 10000 correct (76.07)
Epoch 60, Iteration 0, loss = 0.0005
Epoch 60, Iteration 100, loss = 0.0002
Epoch 60, Iteration 200, loss = 0.0005
Epoch 60, Iteration 300, loss = 0.0003
Eval 7613 / 10000 correct (76.13)
Epoch 61, Iteration 0, loss = 0.0001
Epoch 61, Iteration 100, loss = 0.0002
Epoch 61, Iteration 200, loss = 0.0001
Epoch 61, Iteration 300, loss = 0.0004
Eval 7627 / 10000 correct (76.27)
Epoch 62, Iteration 0, loss = 0.0003
Epoch 62, Iteration 100, loss = 0.0003
Epoch 62, Iteration 200, loss = 0.0001
Epoch 62, Iteration 300, loss = 0.0005
Eval 7619 / 10000 correct (76.19)
Epoch 63, Iteration 0, loss = 0.0004
Epoch 63, Iteration 100, loss = 0.0001
Epoch 63, Iteration 200, loss = 0.0002
Epoch 63, Iteration 300, loss = 0.0000
Eval 7607 / 10000 correct (76.07)
Epoch 64, Iteration 0, loss = 0.0003
Epoch 64, Iteration 100, loss = 0.0004
Epoch 64, Iteration 200, loss = 0.0001
Epoch 64, Iteration 300, loss = 0.0006
Eval 7609 / 10000 correct (76.09)
Epoch 65, Iteration 0, loss = 0.0002
```

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Epoch 65, Iteration 100, loss = 0.0002
Epoch 65, Iteration 200, loss = 0.0001
Epoch 65, Iteration 300, loss = 0.0001
Eval 7610 / 10000 correct (76.10)
Epoch 66, Iteration 0, loss = 0.0000
Epoch 66, Iteration 100, loss = 0.0002
Epoch 66, Iteration 200, loss = 0.0001
Epoch 66, Iteration 300, loss = 0.0003
Eval 7616 / 10000 correct (76.16)
Epoch 67, Iteration 0, loss = 0.0001
Epoch 67, Iteration 100, loss = 0.0000
Epoch 67, Iteration 200, loss = 0.0001
Epoch 67, Iteration 300, loss = 0.0001
Eval 7618 / 10000 correct (76.18)
Epoch 68, Iteration 0, loss = 0.0001
Epoch 68, Iteration 100, loss = 0.0001
Epoch 68, Iteration 200, loss = 0.0001
Epoch 68, Iteration 300, loss = 0.0002
Eval 7622 / 10000 correct (76.22)
Epoch 69, Iteration 0, loss = 0.0000
Epoch 69, Iteration 100, loss = 0.0002
Epoch 69, Iteration 200, loss = 0.0001
Epoch 69, Iteration 300, loss = 0.0001
Eval 7621 / 10000 correct (76.21)
Epoch 70, Iteration 0, loss = 0.0000
Epoch 70, Iteration 100, loss = 0.0001
Epoch 70, Iteration 200, loss = 0.0002
Epoch 70, Iteration 300, loss = 0.0001
Eval 7622 / 10000 correct (76.22)
Epoch 71, Iteration 0, loss = 0.0001
Epoch 71, Iteration 100, loss = 0.0001
Epoch 71, Iteration 200, loss = 0.0001
Epoch 71, Iteration 300, loss = 0.0000
Eval 7631 / 10000 correct (76.31)
Epoch 72, Iteration 0, loss = 0.0001
Epoch 72, Iteration 100, loss = 0.0000
Epoch 72, Iteration 200, loss = 0.0001
Epoch 72, Iteration 300, loss = 0.0000
Eval 7621 / 10000 correct (76.21)
Epoch 73, Iteration 0, loss = 0.0001
Epoch 73, Iteration 100, loss = 0.0001
Epoch 73, Iteration 200, loss = 0.0001
Epoch 73, Iteration 300, loss = 0.0001
Eval 7631 / 10000 correct (76.31)
Epoch 74, Iteration 0, loss = 0.0000
Epoch 74, Iteration 100, loss = 0.0000
Epoch 74, Iteration 200, loss = 0.0001
Epoch 74, Iteration 300, loss = 0.0001
Eval 7633 / 10000 correct (76.33)
Epoch 75, Iteration 0, loss = 0.0000
Epoch 75, Iteration 100, loss = 0.0001
Epoch 75, Iteration 200, loss = 0.0001
Epoch 75, Iteration 300, loss = 0.0001
Eval 7624 / 10000 correct (76.24)
Epoch 76, Iteration 0, loss = 0.0000
```

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Epoch 76, Iteration 100, loss = 0.0000
Epoch 76, Iteration 200, loss = 0.0001
Epoch 76, Iteration 300, loss = 0.0001
Eval 7631 / 10000 correct (76.31)
Epoch 77, Iteration 0, loss = 0.0001
Epoch 77, Iteration 100, loss = 0.0001
Epoch 77, Iteration 200, loss = 0.0001
Epoch 77, Iteration 300, loss = 0.0001
Eval 7627 / 10000 correct (76.27)
Epoch 78, Iteration 0, loss = 0.0000
Epoch 78, Iteration 100, loss = 0.0000
Epoch 78, Iteration 200, loss = 0.0001
Epoch 78, Iteration 300, loss = 0.0001
Eval 7628 / 10000 correct (76.28)
Epoch 79, Iteration 0, loss = 0.0001
Epoch 79, Iteration 100, loss = 0.0000
Epoch 79, Iteration 200, loss = 0.0000
Epoch 79, Iteration 300, loss = 0.0001
Eval 7626 / 10000 correct (76.26)
Epoch 80, Iteration 0, loss = 0.0001
Epoch 80, Iteration 100, loss = 0.0001
Epoch 80, Iteration 200, loss = 0.0001
Epoch 80, Iteration 300, loss = 0.0001
Eval 7632 / 10000 correct (76.32)
Epoch 81, Iteration 0, loss = 0.0000
Epoch 81, Iteration 100, loss = 0.0000
Epoch 81, Iteration 200, loss = 0.0002
Epoch 81, Iteration 300, loss = 0.0001
Eval 7629 / 10000 correct (76.29)
Epoch 82, Iteration 0, loss = 0.0000
Epoch 82, Iteration 100, loss = 0.0000
Epoch 82, Iteration 200, loss = 0.0001
Epoch 82, Iteration 300, loss = 0.0001
Eval 7638 / 10000 correct (76.38)
Epoch 83, Iteration 0, loss = 0.0000
Epoch 83, Iteration 100, loss = 0.0001
Epoch 83, Iteration 200, loss = 0.0001
Epoch 83, Iteration 300, loss = 0.0000
Eval 7633 / 10000 correct (76.33)
Epoch 84, Iteration 0, loss = 0.0000
Epoch 84, Iteration 100, loss = 0.0000
Epoch 84, Iteration 200, loss = 0.0000
Epoch 84, Iteration 300, loss = 0.0001
Eval 7637 / 10000 correct (76.37)
Epoch 85, Iteration 0, loss = 0.0000
Epoch 85, Iteration 100, loss = 0.0000
Epoch 85, Iteration 200, loss = 0.0001
Epoch 85, Iteration 300, loss = 0.0000
Eval 7634 / 10000 correct (76.34)
Epoch 86, Iteration 0, loss = 0.0001
Epoch 86, Iteration 100, loss = 0.0002
Epoch 86, Iteration 200, loss = 0.0001
Epoch 86, Iteration 300, loss = 0.0000
Eval 7639 / 10000 correct (76.39)
Epoch 87, Iteration 0, loss = 0.0000
```

```
Epoch 87, Iteration 100, loss = 0.0000
Epoch 87, Iteration 200, loss = 0.0002
Epoch 87, Iteration 300, loss = 0.0001
Eval 7630 / 10000 correct (76.30)
Epoch 88, Iteration 0, loss = 0.0001
Epoch 88, Iteration 100, loss = 0.0000
Epoch 88, Iteration 200, loss = 0.0000
Epoch 88, Iteration 300, loss = 0.0000
Eval 7633 / 10000 correct (76.33)
Epoch 89, Iteration 0, loss = 0.0001
Epoch 89, Iteration 100, loss = 0.0001
Epoch 89, Iteration 200, loss = 0.0000
Epoch 89, Iteration 300, loss = 0.0004
Eval 7629 / 10000 correct (76.29)
Epoch 90, Iteration 0, loss = 0.0000
Epoch 90, Iteration 100, loss = 0.0002
Epoch 90, Iteration 200, loss = 0.0001
Epoch 90, Iteration 300, loss = 0.0000
Eval 7628 / 10000 correct (76.28)
Epoch 91, Iteration 0, loss = 0.0001
Epoch 91, Iteration 100, loss = 0.0000
Epoch 91, Iteration 200, loss = 0.0001
Epoch 91, Iteration 300, loss = 0.0000
Eval 7642 / 10000 correct (76.42)
Epoch 92, Iteration 0, loss = 0.0002
Epoch 92, Iteration 100, loss = 0.0001
Epoch 92, Iteration 200, loss = 0.0001
Epoch 92, Iteration 300, loss = 0.0001
Eval 7634 / 10000 correct (76.34)
Epoch 93, Iteration 0, loss = 0.0000
Epoch 93, Iteration 100, loss = 0.0001
Epoch 93, Iteration 200, loss = 0.0000
Epoch 93, Iteration 300, loss = 0.0001
Eval 7633 / 10000 correct (76.33)
Epoch 94, Iteration 0, loss = 0.0001
Epoch 94, Iteration 100, loss = 0.0000
Epoch 94, Iteration 200, loss = 0.0000
Epoch 94, Iteration 300, loss = 0.0000
Eval 7625 / 10000 correct (76.25)
Epoch 95, Iteration 0, loss = 0.0001
Epoch 95, Iteration 100, loss = 0.0000
Epoch 95, Iteration 200, loss = 0.0001
Epoch 95, Iteration 300, loss = 0.0001
Eval 7630 / 10000 correct (76.30)
Epoch 96, Iteration 0, loss = 0.0001
Epoch 96, Iteration 100, loss = 0.0000
Epoch 96, Iteration 200, loss = 0.0000
Epoch 96, Iteration 300, loss = 0.0000
Eval 7629 / 10000 correct (76.29)
Epoch 97, Iteration 0, loss = 0.0001
Epoch 97, Iteration 100, loss = 0.0000
Epoch 97, Iteration 200, loss = 0.0000
Epoch 97, Iteration 300, loss = 0.0000
Eval 7634 / 10000 correct (76.34)
Epoch 98, Iteration 0, loss = 0.0000
```

```
Epoch 98, Iteration 100, loss = 0.0000
Epoch 98, Iteration 200, loss = 0.0000
Epoch 98, Iteration 300, loss = 0.0000
Eval 7636 / 10000 correct (76.36)
Epoch 99, Iteration 0, loss = 0.0000
Epoch 99, Iteration 100, loss = 0.0000
Epoch 99, Iteration 200, loss = 0.0000
Epoch 99, Iteration 300, loss = 0.0000
Eval 7634 / 10000 correct (76.34)
```

Tab 3

MNIST training output

```
Epoch 0, Iteration 0, loss = 2.2964
Epoch 0, Iteration 100, loss = 0.4279
Epoch 0, Iteration 200, loss = 0.2643
Epoch 0, Iteration 300, loss = 0.2244
Epoch 0, Iteration 400, loss = 0.1880
Eval 9415 / 10000 correct (94.15)
Epoch 1, Iteration 0, loss = 0.2398
Epoch 1, Iteration 100, loss = 0.2263
Epoch 1, Iteration 200, loss = 0.1790
Epoch 1, Iteration 300, loss = 0.1663
Epoch 1, Iteration 400, loss = 0.1186
Eval 9572 / 10000 correct (95.72)
Epoch 2, Iteration 0, loss = 0.1595
Epoch 2, Iteration 100, loss = 0.0701
Epoch 2, Iteration 200, loss = 0.1148
Epoch 2, Iteration 300, loss = 0.1091
Epoch 2, Iteration 400, loss = 0.1122
Eval 9695 / 10000 correct (96.95)
Epoch 3, Iteration 0, loss = 0.0636
Epoch 3, Iteration 100, loss = 0.0736
Epoch 3, Iteration 200, loss = 0.1786
Epoch 3, Iteration 300, loss = 0.1464
Epoch 3, Iteration 400, loss = 0.0662
Eval 9719 / 10000 correct (97.19)
Epoch 4, Iteration 0, loss = 0.0515
Epoch 4, Iteration 100, loss = 0.3093
Epoch 4, Iteration 200, loss = 0.0237
Epoch 4, Iteration 300, loss = 0.1166
Epoch 4, Iteration 400, loss = 0.0704
Eval 9705 / 10000 correct (97.05)
Epoch 5, Iteration 0, loss = 0.0716
Epoch 5, Iteration 100, loss = 0.0802
Epoch 5, Iteration 200, loss = 0.0690
Epoch 5, Iteration 300, loss = 0.0934
Epoch 5, Iteration 400, loss = 0.0884
Eval 9739 / 10000 correct (97.39)
Epoch 6, Iteration 0, loss = 0.0407
Epoch 6, Iteration 100, loss = 0.0743
Epoch 6, Iteration 200, loss = 0.0866
Epoch 6, Iteration 300, loss = 0.0524
Epoch 6, Iteration 400, loss = 0.0550
Eval 9772 / 10000 correct (97.72)
Epoch 7, Iteration 0, loss = 0.0598
Epoch 7, Iteration 100, loss = 0.0434
Epoch 7, Iteration 200, loss = 0.0704
Epoch 7, Iteration 300, loss = 0.0629
Epoch 7, Iteration 400, loss = 0.0156
Eval 9772 / 10000 correct (97.72)
Epoch 8, Iteration 0, loss = 0.0786
Epoch 8, Iteration 100, loss = 0.0421
Epoch 8, Iteration 200, loss = 0.1566
Epoch 8, Iteration 300, loss = 0.0215
Epoch 8, Iteration 400, loss = 0.0295
Eval 9747 / 10000 correct (97.47)
```

```
Epoch 9, Iteration 0, loss = 0.0415
Epoch 9, Iteration 100, loss = 0.0317
Epoch 9, Iteration 200, loss = 0.0640
Epoch 9, Iteration 300, loss = 0.0703
Epoch 9, Iteration 400, loss = 0.0639
Eval 9794 / 10000 correct (97.94)
Epoch 10, Iteration 0, loss = 0.0135
Epoch 10, Iteration 100, loss = 0.0249
Epoch 10, Iteration 200, loss = 0.0577
Epoch 10, Iteration 300, loss = 0.0154
Epoch 10, Iteration 400, loss = 0.0441
Eval 9800 / 10000 correct (98.00)
Epoch 11, Iteration 0, loss = 0.0364
Epoch 11, Iteration 100, loss = 0.0142
Epoch 11, Iteration 200, loss = 0.0348
Epoch 11, Iteration 300, loss = 0.0206
Epoch 11, Iteration 400, loss = 0.0612
Eval 9821 / 10000 correct (98.21)
Epoch 12, Iteration 0, loss = 0.0080
Epoch 12, Iteration 100, loss = 0.0308
Epoch 12, Iteration 200, loss = 0.0086
Epoch 12, Iteration 300, loss = 0.0146
Epoch 12, Iteration 400, loss = 0.0094
Eval 9816 / 10000 correct (98.16)
Epoch 13, Iteration 0, loss = 0.0693
Epoch 13, Iteration 100, loss = 0.0326
Epoch 13, Iteration 200, loss = 0.0255
Epoch 13, Iteration 300, loss = 0.0230
Epoch 13, Iteration 400, loss = 0.0488
Eval 9799 / 10000 correct (97.99)
Epoch 14, Iteration 0, loss = 0.0191
Epoch 14, Iteration 100, loss = 0.0262
Epoch 14, Iteration 200, loss = 0.1027
Epoch 14, Iteration 300, loss = 0.0264
Epoch 14, Iteration 400, loss = 0.0338
Eval 9804 / 10000 correct (98.04)
Epoch 15, Iteration 0, loss = 0.0468
Epoch 15, Iteration 100, loss = 0.0359
Epoch 15, Iteration 200, loss = 0.0102
Epoch 15, Iteration 300, loss = 0.0254
Epoch 15, Iteration 400, loss = 0.0234
Eval 9800 / 10000 correct (98.00)
Epoch 16, Iteration 0, loss = 0.0559
Epoch 16, Iteration 100, loss = 0.0130
Epoch 16, Iteration 200, loss = 0.0091
Epoch 16, Iteration 300, loss = 0.0063
Epoch 16, Iteration 400, loss = 0.0591
Eval 9816 / 10000 correct (98.16)
Epoch 17, Iteration 0, loss = 0.0219
Epoch 17, Iteration 100, loss = 0.0224
Epoch 17, Iteration 200, loss = 0.0204
Epoch 17, Iteration 300, loss = 0.0238
Epoch 17, Iteration 400, loss = 0.0246
Eval 9825 / 10000 correct (98.25)
Epoch 18, Iteration 0, loss = 0.0234
```

```
Epoch 18, Iteration 100, loss = 0.0098
Epoch 18, Iteration 200, loss = 0.0163
Epoch 18, Iteration 300, loss = 0.0103
Epoch 18, Iteration 400, loss = 0.0397
Eval 9843 / 10000 correct (98.43)
Epoch 19, Iteration 0, loss = 0.0279
Epoch 19, Iteration 100, loss = 0.0109
Epoch 19, Iteration 200, loss = 0.0129
Epoch 19, Iteration 300, loss = 0.0141
Epoch 19, Iteration 400, loss = 0.0108
Eval 9833 / 10000 correct (98.33)
Epoch 20, Iteration 0, loss = 0.0053
Epoch 20, Iteration 100, loss = 0.0074
Epoch 20, Iteration 200, loss = 0.0066
Epoch 20, Iteration 300, loss = 0.0099
Epoch 20, Iteration 400, loss = 0.0081
Eval 9833 / 10000 correct (98.33)
Epoch 21, Iteration 0, loss = 0.0055
Epoch 21, Iteration 100, loss = 0.0090
Epoch 21, Iteration 200, loss = 0.0095
Epoch 21, Iteration 300, loss = 0.0059
Epoch 21, Iteration 400, loss = 0.0066
Eval 9816 / 10000 correct (98.16)
Epoch 22, Iteration 0, loss = 0.0187
Epoch 22, Iteration 100, loss = 0.0059
Epoch 22, Iteration 200, loss = 0.0022
Epoch 22, Iteration 300, loss = 0.0037
Epoch 22, Iteration 400, loss = 0.0048
Eval 9842 / 10000 correct (98.42)
Epoch 23, Iteration 0, loss = 0.0046
Epoch 23, Iteration 100, loss = 0.0113
Epoch 23, Iteration 200, loss = 0.0058
Epoch 23, Iteration 300, loss = 0.0017
Epoch 23, Iteration 400, loss = 0.0060
Eval 9848 / 10000 correct (98.48)
Epoch 24, Iteration 0, loss = 0.0068
Epoch 24, Iteration 100, loss = 0.0032
Epoch 24, Iteration 200, loss = 0.0036
Epoch 24, Iteration 300, loss = 0.0093
Epoch 24, Iteration 400, loss = 0.0079
Eval 9839 / 10000 correct (98.39)
Epoch 25, Iteration 0, loss = 0.0100
Epoch 25, Iteration 100, loss = 0.0007
Epoch 25, Iteration 200, loss = 0.0038
Epoch 25, Iteration 300, loss = 0.0104
Epoch 25, Iteration 400, loss = 0.0026
Eval 9843 / 10000 correct (98.43)
Epoch 26, Iteration 0, loss = 0.0017
Epoch 26, Iteration 100, loss = 0.0046
Epoch 26, Iteration 200, loss = 0.0027
Epoch 26, Iteration 300, loss = 0.0109
Epoch 26, Iteration 400, loss = 0.0023
Eval 9831 / 10000 correct (98.31)
Epoch 27, Iteration 0, loss = 0.0227
Epoch 27, Iteration 100, loss = 0.0013
```

```
Epoch 27, Iteration 200, loss = 0.0077
Epoch 27, Iteration 300, loss = 0.0034
Epoch 27, Iteration 400, loss = 0.0081
Eval 9829 / 10000 correct (98.29)
Epoch 28, Iteration 0, loss = 0.0066
Epoch 28, Iteration 100, loss = 0.0029
Epoch 28, Iteration 200, loss = 0.0019
Epoch 28, Iteration 300, loss = 0.0045
Epoch 28, Iteration 400, loss = 0.0054
Eval 9846 / 10000 correct (98.46)
Epoch 29, Iteration 0, loss = 0.0012
Epoch 29, Iteration 100, loss = 0.0041
Epoch 29, Iteration 200, loss = 0.0082
Epoch 29, Iteration 300, loss = 0.0026
Epoch 29, Iteration 400, loss = 0.0013
Eval 9854 / 10000 correct (98.54)
Epoch 30, Iteration 0, loss = 0.0020
Epoch 30, Iteration 100, loss = 0.0043
Epoch 30, Iteration 200, loss = 0.0016
Epoch 30, Iteration 300, loss = 0.0025
Epoch 30, Iteration 400, loss = 0.0023
Eval 9842 / 10000 correct (98.42)
Epoch 31, Iteration 0, loss = 0.0014
Epoch 31, Iteration 100, loss = 0.0046
Epoch 31, Iteration 200, loss = 0.0037
Epoch 31, Iteration 300, loss = 0.0024
Epoch 31, Iteration 400, loss = 0.0053
Eval 9851 / 10000 correct (98.51)
Epoch 32, Iteration 0, loss = 0.0055
Epoch 32, Iteration 100, loss = 0.0024
Epoch 32, Iteration 200, loss = 0.0041
Epoch 32, Iteration 300, loss = 0.0025
Epoch 32, Iteration 400, loss = 0.0019
Eval 9843 / 10000 correct (98.43)
Epoch 33, Iteration 0, loss = 0.0032
Epoch 33, Iteration 100, loss = 0.0005
Epoch 33, Iteration 200, loss = 0.0031
Epoch 33, Iteration 300, loss = 0.0017
Epoch 33, Iteration 400, loss = 0.0033
Eval 9855 / 10000 correct (98.55)
Epoch 34, Iteration 0, loss = 0.0028
Epoch 34, Iteration 100, loss = 0.0014
Epoch 34, Iteration 200, loss = 0.0013
Epoch 34, Iteration 300, loss = 0.0028
Epoch 34, Iteration 400, loss = 0.0010
Eval 9850 / 10000 correct (98.50)
Epoch 35, Iteration 0, loss = 0.0008
Epoch 35, Iteration 100, loss = 0.0007
Epoch 35, Iteration 200, loss = 0.0006
Epoch 35, Iteration 300, loss = 0.0008
Epoch 35, Iteration 400, loss = 0.0011
Eval 9846 / 10000 correct (98.46)
Epoch 36, Iteration 0, loss = 0.0010
Epoch 36, Iteration 100, loss = 0.0025
Epoch 36, Iteration 200, loss = 0.0009
```

```
Epoch 36, Iteration 300, loss = 0.0011
Epoch 36, Iteration 400, loss = 0.0026
Eval 9856 / 10000 correct (98.56)
Epoch 37, Iteration 0, loss = 0.0017
Epoch 37, Iteration 100, loss = 0.0011
Epoch 37, Iteration 200, loss = 0.0007
Epoch 37, Iteration 300, loss = 0.0017
Epoch 37, Iteration 400, loss = 0.0007
Eval 9850 / 10000 correct (98.50)
Epoch 38, Iteration 0, loss = 0.0002
Epoch 38, Iteration 100, loss = 0.0013
Epoch 38, Iteration 200, loss = 0.0014
Epoch 38, Iteration 300, loss = 0.0004
Epoch 38, Iteration 400, loss = 0.0008
Eval 9854 / 10000 correct (98.54)
Epoch 39, Iteration 0, loss = 0.0019
Epoch 39, Iteration 100, loss = 0.0025
Epoch 39, Iteration 200, loss = 0.0015
Epoch 39, Iteration 300, loss = 0.0001
Epoch 39, Iteration 400, loss = 0.0002
Eval 9848 / 10000 correct (98.48)
Epoch 40, Iteration 0, loss = 0.0025
Epoch 40, Iteration 100, loss = 0.0001
Epoch 40, Iteration 200, loss = 0.0012
Epoch 40, Iteration 300, loss = 0.0033
Epoch 40, Iteration 400, loss = 0.0024
Eval 9849 / 10000 correct (98.49)
Epoch 41, Iteration 0, loss = 0.0005
Epoch 41, Iteration 100, loss = 0.0013
Epoch 41, Iteration 200, loss = 0.0009
Epoch 41, Iteration 300, loss = 0.0008
Epoch 41, Iteration 400, loss = 0.0027
Eval 9851 / 10000 correct (98.51)
Epoch 42, Iteration 0, loss = 0.0005
Epoch 42, Iteration 100, loss = 0.0008
Epoch 42, Iteration 200, loss = 0.0000
Epoch 42, Iteration 300, loss = 0.0015
Epoch 42, Iteration 400, loss = 0.0005
Eval 9854 / 10000 correct (98.54)
Epoch 43, Iteration 0, loss = 0.0010
Epoch 43, Iteration 100, loss = 0.0015
Epoch 43, Iteration 200, loss = 0.0006
Epoch 43, Iteration 300, loss = 0.0016
Epoch 43, Iteration 400, loss = 0.0004
Eval 9850 / 10000 correct (98.50)
Epoch 44, Iteration 0, loss = 0.0003
Epoch 44, Iteration 100, loss = 0.0005
Epoch 44, Iteration 200, loss = 0.0016
Epoch 44, Iteration 300, loss = 0.0005
Epoch 44, Iteration 400, loss = 0.0004
Eval 9858 / 10000 correct (98.58)
Epoch 45, Iteration 0, loss = 0.0009
Epoch 45, Iteration 100, loss = 0.0008
Epoch 45, Iteration 200, loss = 0.0017
Epoch 45, Iteration 300, loss = 0.0006
```

```
Epoch 45, Iteration 400, loss = 0.0017
Eval 9853 / 10000 correct (98.53)
Epoch 46, Iteration 0, loss = 0.0006
Epoch 46, Iteration 100, loss = 0.0009
Epoch 46, Iteration 200, loss = 0.0010
Epoch 46, Iteration 300, loss = 0.0010
Epoch 46, Iteration 400, loss = 0.0011
Eval 9850 / 10000 correct (98.50)
Epoch 47, Iteration 0, loss = 0.0002
Epoch 47, Iteration 100, loss = 0.0005
Epoch 47, Iteration 200, loss = 0.0007
Epoch 47, Iteration 300, loss = 0.0008
Epoch 47, Iteration 400, loss = 0.0025
Eval 9853 / 10000 correct (98.53)
Epoch 48, Iteration 0, loss = 0.0018
Epoch 48, Iteration 100, loss = 0.0003
Epoch 48, Iteration 200, loss = 0.0002
Epoch 48, Iteration 300, loss = 0.0010
Epoch 48, Iteration 400, loss = 0.0003
Eval 9853 / 10000 correct (98.53)
Epoch 49, Iteration 0, loss = 0.0009
Epoch 49, Iteration 100, loss = 0.0003
Epoch 49, Iteration 200, loss = 0.0004
Epoch 49, Iteration 300, loss = 0.0004
Epoch 49, Iteration 400, loss = 0.0014
Eval 9851 / 10000 correct (98.51)
Epoch 50, Iteration 0, loss = 0.0019
Epoch 50, Iteration 100, loss = 0.0005
Epoch 50, Iteration 200, loss = 0.0012
Epoch 50, Iteration 300, loss = 0.0006
Epoch 50, Iteration 400, loss = 0.0001
Eval 9852 / 10000 correct (98.52)
Epoch 51, Iteration 0, loss = 0.0004
Epoch 51, Iteration 100, loss = 0.0005
Epoch 51, Iteration 200, loss = 0.0004
Epoch 51, Iteration 300, loss = 0.0009
Epoch 51, Iteration 400, loss = 0.0002
Eval 9851 / 10000 correct (98.51)
Epoch 52, Iteration 0, loss = 0.0004
Epoch 52, Iteration 100, loss = 0.0011
Epoch 52, Iteration 200, loss = 0.0002
Epoch 52, Iteration 300, loss = 0.0005
Epoch 52, Iteration 400, loss = 0.0002
Eval 9850 / 10000 correct (98.50)
Epoch 53, Iteration 0, loss = 0.0005
Epoch 53, Iteration 100, loss = 0.0007
Epoch 53, Iteration 200, loss = 0.0007
Epoch 53, Iteration 300, loss = 0.0004
Epoch 53, Iteration 400, loss = 0.0009
Eval 9854 / 10000 correct (98.54)
Epoch 54, Iteration 0, loss = 0.0006
Epoch 54, Iteration 100, loss = 0.0005
Epoch 54, Iteration 200, loss = 0.0003
Epoch 54, Iteration 300, loss = 0.0004
Epoch 54, Iteration 400, loss = 0.0003
```

```
Eval 9854 / 10000 correct (98.54)
Epoch 55, Iteration 0, loss = 0.0006
Epoch 55, Iteration 100, loss = 0.0007
Epoch 55, Iteration 200, loss = 0.0006
Epoch 55, Iteration 300, loss = 0.0002
Epoch 55, Iteration 400, loss = 0.0002
Eval 9853 / 10000 correct (98.53)
Epoch 56, Iteration 0, loss = 0.0002
Epoch 56, Iteration 100, loss = 0.0003
Epoch 56, Iteration 200, loss = 0.0015
Epoch 56, Iteration 300, loss = 0.0007
Epoch 56, Iteration 400, loss = 0.0003
Eval 9852 / 10000 correct (98.52)
Epoch 57, Iteration 0, loss = 0.0004
Epoch 57, Iteration 100, loss = 0.0006
Epoch 57, Iteration 200, loss = 0.0008
Epoch 57, Iteration 300, loss = 0.0003
Epoch 57, Iteration 400, loss = 0.0004
Eval 9855 / 10000 correct (98.55)
Epoch 58, Iteration 0, loss = 0.0004
Epoch 58, Iteration 100, loss = 0.0004
Epoch 58, Iteration 200, loss = 0.0009
Epoch 58, Iteration 300, loss = 0.0006
Epoch 58, Iteration 400, loss = 0.0005
Eval 9852 / 10000 correct (98.52)
Epoch 59, Iteration 0, loss = 0.0003
Epoch 59, Iteration 100, loss = 0.0004
Epoch 59, Iteration 200, loss = 0.0007
Epoch 59, Iteration 300, loss = 0.0006
Epoch 59, Iteration 400, loss = 0.0006
Eval 9852 / 10000 correct (98.52)
Epoch 60, Iteration 0, loss = 0.0002
Epoch 60, Iteration 100, loss = 0.0003
Epoch 60, Iteration 200, loss = 0.0007
Epoch 60, Iteration 300, loss = 0.0005
Epoch 60, Iteration 400, loss = 0.0002
Eval 9853 / 10000 correct (98.53)
Epoch 61, Iteration 0, loss = 0.0002
Epoch 61, Iteration 100, loss = 0.0004
Epoch 61, Iteration 200, loss = 0.0002
Epoch 61, Iteration 300, loss = 0.0004
Epoch 61, Iteration 400, loss = 0.0007
Eval 9853 / 10000 correct (98.53)
Epoch 62, Iteration 0, loss = 0.0001
Epoch 62, Iteration 100, loss = 0.0005
Epoch 62, Iteration 200, loss = 0.0013
Epoch 62, Iteration 300, loss = 0.0002
Epoch 62, Iteration 400, loss = 0.0001
Eval 9856 / 10000 correct (98.56)
Epoch 63, Iteration 0, loss = 0.0005
Epoch 63, Iteration 100, loss = 0.0003
Epoch 63, Iteration 200, loss = 0.0005
Epoch 63, Iteration 300, loss = 0.0007
Epoch 63, Iteration 400, loss = 0.0002
Eval 9856 / 10000 correct (98.56)
```

```
Epoch 64, Iteration 0, loss = 0.0004
Epoch 64, Iteration 100, loss = 0.0003
Epoch 64, Iteration 200, loss = 0.0002
Epoch 64, Iteration 300, loss = 0.0003
Epoch 64, Iteration 400, loss = 0.0008
Eval 9852 / 10000 correct (98.52)
Epoch 65, Iteration 0, loss = 0.0003
Epoch 65, Iteration 100, loss = 0.0003
Epoch 65, Iteration 200, loss = 0.0004
Epoch 65, Iteration 300, loss = 0.0004
Epoch 65, Iteration 400, loss = 0.0006
Eval 9852 / 10000 correct (98.52)
Epoch 66, Iteration 0, loss = 0.0003
Epoch 66, Iteration 100, loss = 0.0007
Epoch 66, Iteration 200, loss = 0.0004
Epoch 66, Iteration 300, loss = 0.0001
Epoch 66, Iteration 400, loss = 0.0005
Eval 9855 / 10000 correct (98.55)
Epoch 67, Iteration 0, loss = 0.0001
Epoch 67, Iteration 100, loss = 0.0006
Epoch 67, Iteration 200, loss = 0.0001
Epoch 67, Iteration 300, loss = 0.0001
Epoch 67, Iteration 400, loss = 0.0002
Eval 9853 / 10000 correct (98.53)
Epoch 68, Iteration 0, loss = 0.0007
Epoch 68, Iteration 100, loss = 0.0003
Epoch 68, Iteration 200, loss = 0.0007
Epoch 68, Iteration 300, loss = 0.0004
Epoch 68, Iteration 400, loss = 0.0010
Eval 9853 / 10000 correct (98.53)
Epoch 69, Iteration 0, loss = 0.0003
Epoch 69, Iteration 100, loss = 0.0001
Epoch 69, Iteration 200, loss = 0.0006
Epoch 69, Iteration 300, loss = 0.0007
Epoch 69, Iteration 400, loss = 0.0004
Eval 9855 / 10000 correct (98.55)
Epoch 70, Iteration 0, loss = 0.0004
Epoch 70, Iteration 100, loss = 0.0004
Epoch 70, Iteration 200, loss = 0.0003
Epoch 70, Iteration 300, loss = 0.0002
Epoch 70, Iteration 400, loss = 0.0002
Eval 9851 / 10000 correct (98.51)
Epoch 71, Iteration 0, loss = 0.0010
Epoch 71, Iteration 100, loss = 0.0003
Epoch 71, Iteration 200, loss = 0.0002
Epoch 71, Iteration 300, loss = 0.0001
Epoch 71, Iteration 400, loss = 0.0003
Eval 9854 / 10000 correct (98.54)
Epoch 72, Iteration 0, loss = 0.0003
Epoch 72, Iteration 100, loss = 0.0001
Epoch 72, Iteration 200, loss = 0.0002
Epoch 72, Iteration 300, loss = 0.0002
Epoch 72, Iteration 400, loss = 0.0001
Eval 9851 / 10000 correct (98.51)
Epoch 73, Iteration 0, loss = 0.0009
```

```
Epoch 73, Iteration 100, loss = 0.0004
Epoch 73, Iteration 200, loss = 0.0005
Epoch 73, Iteration 300, loss = 0.0002
Epoch 73, Iteration 400, loss = 0.0003
Eval 9852 / 10000 correct (98.52)
Epoch 74, Iteration 0, loss = 0.0004
Epoch 74, Iteration 100, loss = 0.0001
Epoch 74, Iteration 200, loss = 0.0002
Epoch 74, Iteration 300, loss = 0.0001
Epoch 74, Iteration 400, loss = 0.0003
Eval 9851 / 10000 correct (98.51)
Epoch 75, Iteration 0, loss = 0.0005
Epoch 75, Iteration 100, loss = 0.0004
Epoch 75, Iteration 200, loss = 0.0002
Epoch 75, Iteration 300, loss = 0.0001
Epoch 75, Iteration 400, loss = 0.0007
Eval 9858 / 10000 correct (98.58)
Epoch 76, Iteration 0, loss = 0.0005
Epoch 76, Iteration 100, loss = 0.0002
Epoch 76, Iteration 200, loss = 0.0003
Epoch 76, Iteration 300, loss = 0.0007
Epoch 76, Iteration 400, loss = 0.0001
Eval 9849 / 10000 correct (98.49)
Epoch 77, Iteration 0, loss = 0.0009
Epoch 77, Iteration 100, loss = 0.0002
Epoch 77, Iteration 200, loss = 0.0002
Epoch 77, Iteration 300, loss = 0.0008
Epoch 77, Iteration 400, loss = 0.0001
Eval 9852 / 10000 correct (98.52)
Epoch 78, Iteration 0, loss = 0.0003
Epoch 78, Iteration 100, loss = 0.0001
Epoch 78, Iteration 200, loss = 0.0004
Epoch 78, Iteration 300, loss = 0.0003
Epoch 78, Iteration 400, loss = 0.0002
Eval 9850 / 10000 correct (98.50)
Epoch 79, Iteration 0, loss = 0.0002
Epoch 79, Iteration 100, loss = 0.0004
Epoch 79, Iteration 200, loss = 0.0003
Epoch 79, Iteration 300, loss = 0.0001
Epoch 79, Iteration 400, loss = 0.0003
Eval 9851 / 10000 correct (98.51)
Epoch 80, Iteration 0, loss = 0.0004
Epoch 80, Iteration 100, loss = 0.0005
Epoch 80, Iteration 200, loss = 0.0002
Epoch 80, Iteration 300, loss = 0.0005
Epoch 80, Iteration 400, loss = 0.0003
Eval 9854 / 10000 correct (98.54)
Epoch 81, Iteration 0, loss = 0.0002
Epoch 81, Iteration 100, loss = 0.0003
Epoch 81, Iteration 200, loss = 0.0006
Epoch 81, Iteration 300, loss = 0.0004
Epoch 81, Iteration 400, loss = 0.0007
Eval 9851 / 10000 correct (98.51)
Epoch 82, Iteration 0, loss = 0.0002
Epoch 82, Iteration 100, loss = 0.0003
```

```
Epoch 82, Iteration 200, loss = 0.0003
Epoch 82, Iteration 300, loss = 0.0001
Epoch 82, Iteration 400, loss = 0.0001
Eval 9855 / 10000 correct (98.55)
Epoch 83, Iteration 0, loss = 0.0001
Epoch 83, Iteration 100, loss = 0.0002
Epoch 83, Iteration 200, loss = 0.0001
Epoch 83, Iteration 300, loss = 0.0004
Epoch 83, Iteration 400, loss = 0.0002
Eval 9855 / 10000 correct (98.55)
Epoch 84, Iteration 0, loss = 0.0001
Epoch 84, Iteration 100, loss = 0.0002
Epoch 84, Iteration 200, loss = 0.0002
Epoch 84, Iteration 300, loss = 0.0006
Epoch 84, Iteration 400, loss = 0.0003
Eval 9854 / 10000 correct (98.54)
Epoch 85, Iteration 0, loss = 0.0002
Epoch 85, Iteration 100, loss = 0.0002
Epoch 85, Iteration 200, loss = 0.0001
Epoch 85, Iteration 300, loss = 0.0003
Epoch 85, Iteration 400, loss = 0.0002
Eval 9854 / 10000 correct (98.54)
Epoch 86, Iteration 0, loss = 0.0003
Epoch 86, Iteration 100, loss = 0.0001
Epoch 86, Iteration 200, loss = 0.0003
Epoch 86, Iteration 300, loss = 0.0004
Epoch 86, Iteration 400, loss = 0.0001
Eval 9852 / 10000 correct (98.52)
Epoch 87, Iteration 0, loss = 0.0002
Epoch 87, Iteration 100, loss = 0.0002
Epoch 87, Iteration 200, loss = 0.0002
Epoch 87, Iteration 300, loss = 0.0005
Epoch 87, Iteration 400, loss = 0.0004
Eval 9854 / 10000 correct (98.54)
Epoch 88, Iteration 0, loss = 0.0002
Epoch 88, Iteration 100, loss = 0.0002
Epoch 88, Iteration 200, loss = 0.0005
Epoch 88, Iteration 300, loss = 0.0003
Epoch 88, Iteration 400, loss = 0.0003
Eval 9853 / 10000 correct (98.53)
Epoch 89, Iteration 0, loss = 0.0001
Epoch 89, Iteration 100, loss = 0.0003
Epoch 89, Iteration 200, loss = 0.0009
Epoch 89, Iteration 300, loss = 0.0001
Epoch 89, Iteration 400, loss = 0.0002
Eval 9851 / 10000 correct (98.51)
Epoch 90, Iteration 0, loss = 0.0002
Epoch 90, Iteration 100, loss = 0.0001
Epoch 90, Iteration 200, loss = 0.0001
Epoch 90, Iteration 300, loss = 0.0002
Epoch 90, Iteration 400, loss = 0.0006
Eval 9851 / 10000 correct (98.51)
Epoch 91, Iteration 0, loss = 0.0001
Epoch 91, Iteration 100, loss = 0.0001
Epoch 91, Iteration 200, loss = 0.0003
```

```
Epoch 91, Iteration 300, loss = 0.0001
Epoch 91, Iteration 400, loss = 0.0003
Eval 9854 / 10000 correct (98.54)
Epoch 92, Iteration 0, loss = 0.0002
Epoch 92, Iteration 100, loss = 0.0001
Epoch 92, Iteration 200, loss = 0.0003
Epoch 92, Iteration 300, loss = 0.0001
Epoch 92, Iteration 400, loss = 0.0001
Eval 9853 / 10000 correct (98.53)
Epoch 93, Iteration 0, loss = 0.0004
Epoch 93, Iteration 100, loss = 0.0003
Epoch 93, Iteration 200, loss = 0.0002
Epoch 93, Iteration 300, loss = 0.0002
Epoch 93, Iteration 400, loss = 0.0002
Eval 9855 / 10000 correct (98.55)
Epoch 94, Iteration 0, loss = 0.0004
Epoch 94, Iteration 100, loss = 0.0002
Epoch 94, Iteration 200, loss = 0.0002
Epoch 94, Iteration 300, loss = 0.0004
Epoch 94, Iteration 400, loss = 0.0002
Eval 9854 / 10000 correct (98.54)
Epoch 95, Iteration 0, loss = 0.0002
Epoch 95, Iteration 100, loss = 0.0001
Epoch 95, Iteration 200, loss = 0.0003
Epoch 95, Iteration 300, loss = 0.0002
Epoch 95, Iteration 400, loss = 0.0001
Eval 9853 / 10000 correct (98.53)
Epoch 96, Iteration 0, loss = 0.0001
Epoch 96, Iteration 100, loss = 0.0001
Epoch 96, Iteration 200, loss = 0.0001
Epoch 96, Iteration 300, loss = 0.0005
Epoch 96, Iteration 400, loss = 0.0002
Eval 9857 / 10000 correct (98.57)
Epoch 97, Iteration 0, loss = 0.0003
Epoch 97, Iteration 100, loss = 0.0001
Epoch 97, Iteration 200, loss = 0.0002
Epoch 97, Iteration 300, loss = 0.0001
Epoch 97, Iteration 400, loss = 0.0005
Eval 9851 / 10000 correct (98.51)
Epoch 98, Iteration 0, loss = 0.0001
Epoch 98, Iteration 100, loss = 0.0001
Epoch 98, Iteration 200, loss = 0.0002
Epoch 98, Iteration 300, loss = 0.0001
Epoch 98, Iteration 400, loss = 0.0003
Eval 9851 / 10000 correct (98.51)
Epoch 99, Iteration 0, loss = 0.0003
Epoch 99, Iteration 100, loss = 0.0006
Epoch 99, Iteration 200, loss = 0.0005
Epoch 99, Iteration 300, loss = 0.0002
Epoch 99, Iteration 400, loss = 0.0002
Eval 9852 / 10000 correct (98.52)
```

Tab 4

Image Segmentation output

```
Epoch 0, Iteration 0, loss = 3.0949
Epoch 0, Iteration 100, loss = 2.3318
Epoch 0, Iteration 200, loss = 2.2783
Epoch 0, Iteration 300, loss = 2.0712
Epoch 0, Iteration 400, loss = 2.0019
Epoch 0, Iteration 500, loss = 1.8944
Epoch 0, Iteration 600, loss = 1.7630
Epoch 0, Iteration 700, loss = 1.8293
Eval 4253789 / 10240000 correct (41.54)
Epoch 1, Iteration 0, loss = 1.5458
Epoch 1, Iteration 100, loss = 1.7055
Epoch 1, Iteration 200, loss = 1.5809
Epoch 1, Iteration 300, loss = 1.4778
Epoch 1, Iteration 400, loss = 1.5681
Epoch 1, Iteration 500, loss = 1.4123
Epoch 1, Iteration 600, loss = 1.5558
Epoch 1, Iteration 700, loss = 1.3628
Eval 5455754 / 10240000 correct (53.28)
Epoch 2, Iteration 0, loss = 1.2435
Epoch 2, Iteration 100, loss = 1.2227
Epoch 2, Iteration 200, loss = 1.4036
Epoch 2, Iteration 300, loss = 1.2108
Epoch 2, Iteration 400, loss = 1.3614
Epoch 2, Iteration 500, loss = 1.2487
Epoch 2, Iteration 600, loss = 1.0598
Epoch 2, Iteration 700, loss = 1.2898
Eval 5958946 / 10240000 correct (58.19)
Epoch 3, Iteration 0, loss = 1.1516
Epoch 3, Iteration 100, loss = 1.0016
Epoch 3, Iteration 200, loss = 1.0425
Epoch 3, Iteration 300, loss = 0.9481
Epoch 3, Iteration 400, loss = 0.9951
Epoch 3, Iteration 500, loss = 1.0470
Epoch 3, Iteration 600, loss = 1.0682
Epoch 3, Iteration 700, loss = 0.8926
Eval 6209341 / 10240000 correct (60.64)
Epoch 4, Iteration 0, loss = 0.9953
Epoch 4, Iteration 100, loss = 0.8900
Epoch 4, Iteration 200, loss = 0.9884
Epoch 4, Iteration 300, loss = 0.6733
Epoch 4, Iteration 400, loss = 0.8435
Epoch 4, Iteration 500, loss = 0.9017
Epoch 4, Iteration 600, loss = 0.9548
Epoch 4, Iteration 700, loss = 0.8961
Eval 6442347 / 10240000 correct (62.91)
```

```
Epoch 5, Iteration 0, loss = 0.8201
Epoch 5, Iteration 100, loss = 0.8446
Epoch 5, Iteration 200, loss = 0.7500
Epoch 5, Iteration 300, loss = 0.7684
Epoch 5, Iteration 400, loss = 0.5416
Epoch 5, Iteration 500, loss = 0.6341
Epoch 5, Iteration 600, loss = 0.9373
Epoch 5, Iteration 700, loss = 0.7825
Eval 6816560 / 10240000 correct (66.57)
Epoch 6, Iteration 0, loss = 0.6501
Epoch 6, Iteration 100, loss = 0.5633
Epoch 6, Iteration 200, loss = 0.6897
Epoch 6, Iteration 300, loss = 0.5983
Epoch 6, Iteration 400, loss = 0.6889
Epoch 6, Iteration 500, loss = 0.5067
Epoch 6, Iteration 600, loss = 0.6560
Epoch 6, Iteration 700, loss = 0.6136
Eval 6670652 / 10240000 correct (65.14)
Epoch 7, Iteration 0, loss = 0.4660
Epoch 7, Iteration 100, loss = 0.4937
Epoch 7, Iteration 200, loss = 0.6504
Epoch 7, Iteration 300, loss = 0.5395
Epoch 7, Iteration 400, loss = 0.4818
Epoch 7, Iteration 500, loss = 0.5347
Epoch 7, Iteration 600, loss = 0.6075
Epoch 7, Iteration 700, loss = 0.5613
Eval 6811670 / 10240000 correct (66.52)
Epoch 8, Iteration 0, loss = 0.4162
Epoch 8, Iteration 100, loss = 0.5452
Epoch 8, Iteration 200, loss = 0.2492
Epoch 8, Iteration 300, loss = 0.5143
Epoch 8, Iteration 400, loss = 0.4206
Epoch 8, Iteration 500, loss = 0.7101
Epoch 8, Iteration 600, loss = 0.4786
Epoch 8, Iteration 700, loss = 0.3865
Eval 7072224 / 10240000 correct (69.06)
Epoch 9, Iteration 0, loss = 0.3353
Epoch 9, Iteration 100, loss = 0.2901
Epoch 9, Iteration 200, loss = 0.2600
Epoch 9, Iteration 300, loss = 0.6096
Epoch 9, Iteration 400, loss = 0.4246
Epoch 9, Iteration 500, loss = 0.4902
Epoch 9, Iteration 600, loss = 0.4451
Epoch 9, Iteration 700, loss = 0.4730
Eval 7079877 / 10240000 correct (69.14)
Epoch 10, Iteration 0, loss = 0.3305
Epoch 10, Iteration 100, loss = 0.3325
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Epoch 10, Iteration 200, loss = 0.2417
Epoch 10, Iteration 300, loss = 0.1564
Epoch 10, Iteration 400, loss = 0.2336
Epoch 10, Iteration 500, loss = 0.2341
Epoch 10, Iteration 600, loss = 0.3893
Epoch 10, Iteration 700, loss = 0.5012
Eval 7125616 / 10240000 correct (69.59)
Epoch 11, Iteration 0, loss = 0.1627
Epoch 11, Iteration 100, loss = 0.2718
Epoch 11, Iteration 200, loss = 0.4083
Epoch 11, Iteration 300, loss = 0.2154
Epoch 11, Iteration 400, loss = 0.3685
Epoch 11, Iteration 500, loss = 0.2897
Epoch 11, Iteration 600, loss = 0.1988
Epoch 11, Iteration 700, loss = 0.2761
Eval 7243321 / 10240000 correct (70.74)
Epoch 12, Iteration 0, loss = 0.1614
Epoch 12, Iteration 100, loss = 0.1802
Epoch 12, Iteration 200, loss = 0.2261
Epoch 12, Iteration 300, loss = 0.2611
Epoch 12, Iteration 400, loss = 0.3213
Epoch 12, Iteration 500, loss = 0.1810
Epoch 12, Iteration 600, loss = 0.1512
Epoch 12, Iteration 700, loss = 0.2756
Eval 7213608 / 10240000 correct (70.45)
Epoch 13, Iteration 0, loss = 0.2114
Epoch 13, Iteration 100, loss = 0.1916
Epoch 13, Iteration 200, loss = 0.1362
Epoch 13, Iteration 300, loss = 0.1358
Epoch 13, Iteration 400, loss = 0.1961
Epoch 13, Iteration 500, loss = 0.1196
Epoch 13, Iteration 600, loss = 0.3696
Epoch 13, Iteration 700, loss = 0.3229
Eval 7163921 / 10240000 correct (69.96)
Epoch 14, Iteration 0, loss = 0.1331
Epoch 14, Iteration 100, loss = 0.2344
Epoch 14, Iteration 200, loss = 0.2540
Epoch 14, Iteration 300, loss = 0.2347
Epoch 14, Iteration 400, loss = 0.3506
Epoch 14, Iteration 500, loss = 0.1450
Epoch 14, Iteration 600, loss = 0.2103
Epoch 14, Iteration 700, loss = 0.3332
Eval 7276224 / 10240000 correct (71.06)
Epoch 15, Iteration 0, loss = 0.1678
Epoch 15, Iteration 100, loss = 0.0977
Epoch 15, Iteration 200, loss = 0.1156
Epoch 15, Iteration 300, loss = 0.1555
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Epoch 15, Iteration 400, loss = 0.1008
Epoch 15, Iteration 500, loss = 0.0973
Epoch 15, Iteration 600, loss = 0.1280
Epoch 15, Iteration 700, loss = 0.2522
Eval 7329990 / 10240000 correct (71.58)
Epoch 16, Iteration 0, loss = 0.1246
Epoch 16, Iteration 100, loss = 0.1430
Epoch 16, Iteration 200, loss = 0.1539
Epoch 16, Iteration 300, loss = 0.1125
Epoch 16, Iteration 400, loss = 0.0899
Epoch 16, Iteration 500, loss = 0.2994
Epoch 16, Iteration 600, loss = 0.1961
Epoch 16, Iteration 700, loss = 0.1527
Eval 7257689 / 10240000 correct (70.88)
Epoch 17, Iteration 0, loss = 0.1372
Epoch 17, Iteration 100, loss = 0.0953
Epoch 17, Iteration 200, loss = 0.1193
Epoch 17, Iteration 300, loss = 0.1242
Epoch 17, Iteration 400, loss = 0.1822
Epoch 17, Iteration 500, loss = 0.1787
Epoch 17, Iteration 600, loss = 0.1041
Epoch 17, Iteration 700, loss = 0.1587
Eval 7133195 / 10240000 correct (69.66)
Epoch 18, Iteration 0, loss = 0.1416
Epoch 18, Iteration 100, loss = 0.1052
Epoch 18, Iteration 200, loss = 0.1088
Epoch 18, Iteration 300, loss = 0.2239
Epoch 18, Iteration 400, loss = 0.0696
Epoch 18, Iteration 500, loss = 0.0716
Epoch 18, Iteration 600, loss = 0.0688
Epoch 18, Iteration 700, loss = 0.0747
Eval 7415361 / 10240000 correct (72.42)
Epoch 19, Iteration 0, loss = 0.1786
Epoch 19, Iteration 100, loss = 0.1322
Epoch 19, Iteration 200, loss = 0.0711
Epoch 19, Iteration 300, loss = 0.0663
Epoch 19, Iteration 400, loss = 0.2194
Epoch 19, Iteration 500, loss = 0.2063
Epoch 19, Iteration 600, loss = 0.0629
Epoch 19, Iteration 700, loss = 0.0872
Eval 7257447 / 10240000 correct (70.87)
Epoch 20, Iteration 0, loss = 0.0610
Epoch 20, Iteration 100, loss = 0.0960
Epoch 20, Iteration 200, loss = 0.1247
Epoch 20, Iteration 300, loss = 0.1298
Epoch 20, Iteration 400, loss = 0.0644
Epoch 20, Iteration 500, loss = 0.0916
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Epoch 20, Iteration 600, loss = 0.1860
Epoch 20, Iteration 700, loss = 0.2471
Eval 7238442 / 10240000 correct (70.69)
Epoch 21, Iteration 0, loss = 0.0948
Epoch 21, Iteration 100, loss = 0.0570
Epoch 21, Iteration 200, loss = 0.1281
Epoch 21, Iteration 300, loss = 0.1050
Epoch 21, Iteration 400, loss = 0.1032
Epoch 21, Iteration 500, loss = 0.0836
Epoch 21, Iteration 600, loss = 0.0735
Epoch 21, Iteration 700, loss = 0.1969
Eval 7362931 / 10240000 correct (71.90)
Epoch 22, Iteration 0, loss = 0.0863
Epoch 22, Iteration 100, loss = 0.0831
Epoch 22, Iteration 200, loss = 0.1580
Epoch 22, Iteration 300, loss = 0.1616
Epoch 22, Iteration 400, loss = 0.1521
Epoch 22, Iteration 500, loss = 0.0494
Epoch 22, Iteration 600, loss = 0.1280
Epoch 22, Iteration 700, loss = 0.1437
Eval 7402228 / 10240000 correct (72.29)
Epoch 23, Iteration 0, loss = 0.0594
Epoch 23, Iteration 100, loss = 0.0889
Epoch 23, Iteration 200, loss = 0.1950
Epoch 23, Iteration 300, loss = 0.1353
Epoch 23, Iteration 400, loss = 0.1734
Epoch 23, Iteration 500, loss = 0.0706
Epoch 23, Iteration 600, loss = 0.1291
Epoch 23, Iteration 700, loss = 0.4325
Eval 7446014 / 10240000 correct (72.71)
Epoch 24, Iteration 0, loss = 0.0862
Epoch 24, Iteration 100, loss = 0.1361
Epoch 24, Iteration 200, loss = 0.0475
Epoch 24, Iteration 300, loss = 0.1842
Epoch 24, Iteration 400, loss = 0.2082
Epoch 24, Iteration 500, loss = 0.1543
Epoch 24, Iteration 600, loss = 0.1217
Epoch 24, Iteration 700, loss = 0.1599
Eval 7370968 / 10240000 correct (71.98)
Epoch 25, Iteration 0, loss = 0.0912
Epoch 25, Iteration 100, loss = 0.0706
Epoch 25, Iteration 200, loss = 0.1021
Epoch 25, Iteration 300, loss = 0.1981
Epoch 25, Iteration 400, loss = 0.0728
Epoch 25, Iteration 500, loss = 0.1477
Epoch 25, Iteration 600, loss = 0.0490
Epoch 25, Iteration 700, loss = 0.0793
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Eval 7381788 / 10240000 correct (72.09)
Epoch 26, Iteration 0, loss = 0.1135
Epoch 26, Iteration 100, loss = 0.0395
Epoch 26, Iteration 200, loss = 0.0804
Epoch 26, Iteration 300, loss = 0.0714
Epoch 26, Iteration 400, loss = 0.1065
Epoch 26, Iteration 500, loss = 0.0491
Epoch 26, Iteration 600, loss = 0.0822
Epoch 26, Iteration 700, loss = 0.0867
Eval 7451500 / 10240000 correct (72.77)
Epoch 27, Iteration 0, loss = 0.0390
Epoch 27, Iteration 100, loss = 0.0403
Epoch 27, Iteration 200, loss = 0.1204
Epoch 27, Iteration 300, loss = 0.1753
Epoch 27, Iteration 400, loss = 0.2043
Epoch 27, Iteration 500, loss = 0.1105
Epoch 27, Iteration 600, loss = 0.0933
Epoch 27, Iteration 700, loss = 0.0639
Eval 7394591 / 10240000 correct (72.21)
Epoch 28, Iteration 0, loss = 0.0447
Epoch 28, Iteration 100, loss = 0.0662
Epoch 28, Iteration 200, loss = 0.0322
Epoch 28, Iteration 300, loss = 0.0410
Epoch 28, Iteration 400, loss = 0.1301
Epoch 28, Iteration 500, loss = 0.0827
Epoch 28, Iteration 600, loss = 0.0901
Epoch 28, Iteration 700, loss = 0.0793
Eval 7268575 / 10240000 correct (70.98)
Epoch 29, Iteration 0, loss = 0.1492
Epoch 29, Iteration 100, loss = 0.0810
Epoch 29, Iteration 200, loss = 0.2110
Epoch 29, Iteration 300, loss = 0.0409
Epoch 29, Iteration 400, loss = 0.1078
Epoch 29, Iteration 500, loss = 0.1741
Epoch 29, Iteration 600, loss = 0.2556
Epoch 29, Iteration 700, loss = 0.1494
Eval 7460097 / 10240000 correct (72.85)
Epoch 30, Iteration 0, loss = 0.1090
Epoch 30, Iteration 100, loss = 0.0679
Epoch 30, Iteration 200, loss = 0.0612
Epoch 30, Iteration 300, loss = 0.0507
Epoch 30, Iteration 400, loss = 0.0418
Epoch 30, Iteration 500, loss = 0.0826
Epoch 30, Iteration 600, loss = 0.0924
Epoch 30, Iteration 700, loss = 0.0872
Eval 7562823 / 10240000 correct (73.86)
Epoch 31, Iteration 0, loss = 0.0332
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Epoch 31, Iteration 100, loss = 0.0388
Epoch 31, Iteration 200, loss = 0.0380
Epoch 31, Iteration 300, loss = 0.2032
Epoch 31, Iteration 400, loss = 0.0823
Epoch 31, Iteration 500, loss = 0.0507
Epoch 31, Iteration 600, loss = 0.0658
Epoch 31, Iteration 700, loss = 0.1266
Eval 7380898 / 10240000 correct (72.08)
Epoch 32, Iteration 0, loss = 0.1223
Epoch 32, Iteration 100, loss = 0.1054
Epoch 32, Iteration 200, loss = 0.0424
Epoch 32, Iteration 300, loss = 0.0354
Epoch 32, Iteration 400, loss = 0.0882
Epoch 32, Iteration 500, loss = 0.0967
Epoch 32, Iteration 600, loss = 0.1128
Epoch 32, Iteration 700, loss = 0.0735
Eval 7571875 / 10240000 correct (73.94)
Epoch 33, Iteration 0, loss = 0.0236
Epoch 33, Iteration 100, loss = 0.0364
Epoch 33, Iteration 200, loss = 0.0237
Epoch 33, Iteration 300, loss = 0.0519
Epoch 33, Iteration 400, loss = 0.0274
Epoch 33, Iteration 500, loss = 0.0359
Epoch 33, Iteration 600, loss = 0.0545
Epoch 33, Iteration 700, loss = 0.0419
Eval 7452604 / 10240000 correct (72.78)
Epoch 34, Iteration 0, loss = 0.0844
Epoch 34, Iteration 100, loss = 0.1027
Epoch 34, Iteration 200, loss = 0.0432
Epoch 34, Iteration 300, loss = 0.0352
Epoch 34, Iteration 400, loss = 0.0468
Epoch 34, Iteration 500, loss = 0.0458
Epoch 34, Iteration 600, loss = 0.0497
Epoch 34, Iteration 700, loss = 0.0664
Eval 7290598 / 10240000 correct (71.20)
Epoch 35, Iteration 0, loss = 0.0292
Epoch 35, Iteration 100, loss = 0.0259
Epoch 35, Iteration 200, loss = 0.0297
Epoch 35, Iteration 300, loss = 0.0415
Epoch 35, Iteration 400, loss = 0.0417
Epoch 35, Iteration 500, loss = 0.0378
Epoch 35, Iteration 600, loss = 0.0369
Epoch 35, Iteration 700, loss = 0.0997
Eval 7498775 / 10240000 correct (73.23)
Epoch 36, Iteration 0, loss = 0.0381
Epoch 36, Iteration 100, loss = 0.0749
Epoch 36, Iteration 200, loss = 0.1253
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Epoch 36, Iteration 300, loss = 0.0260
Epoch 36, Iteration 400, loss = 0.0299
Epoch 36, Iteration 500, loss = 0.0262
Epoch 36, Iteration 600, loss = 0.0434
Epoch 36, Iteration 700, loss = 0.0410
Eval 7414635 / 10240000 correct (72.41)
Epoch 37, Iteration 0, loss = 0.0597
Epoch 37, Iteration 100, loss = 0.0478
Epoch 37, Iteration 200, loss = 0.0749
Epoch 37, Iteration 300, loss = 0.0208
Epoch 37, Iteration 400, loss = 0.0463
Epoch 37, Iteration 500, loss = 0.0627
Epoch 37, Iteration 600, loss = 0.0244
Epoch 37, Iteration 700, loss = 0.0993
Eval 7150243 / 10240000 correct (69.83)
Epoch 38, Iteration 0, loss = 0.1091
Epoch 38, Iteration 100, loss = 0.0463
Epoch 38, Iteration 200, loss = 0.0856
Epoch 38, Iteration 300, loss = 0.0419
Epoch 38, Iteration 400, loss = 0.0361
Epoch 38, Iteration 500, loss = 0.0584
Epoch 38, Iteration 600, loss = 0.0950
Epoch 38, Iteration 700, loss = 0.0604
Eval 7427945 / 10240000 correct (72.54)
Epoch 39, Iteration 0, loss = 0.1111
Epoch 39, Iteration 100, loss = 0.0226
Epoch 39, Iteration 200, loss = 0.0713
Epoch 39, Iteration 300, loss = 0.0236
Epoch 39, Iteration 400, loss = 0.0225
Epoch 39, Iteration 500, loss = 0.0344
Epoch 39, Iteration 600, loss = 0.1562
Epoch 39, Iteration 700, loss = 0.0402
Eval 7564855 / 10240000 correct (73.88)
Epoch 40, Iteration 0, loss = 0.0234
Epoch 40, Iteration 100, loss = 0.0586
Epoch 40, Iteration 200, loss = 0.0311
Epoch 40, Iteration 300, loss = 0.0220
Epoch 40, Iteration 400, loss = 0.0308
Epoch 40, Iteration 500, loss = 0.1529
Epoch 40, Iteration 600, loss = 0.0393
Epoch 40, Iteration 700, loss = 0.0329
Eval 7578340 / 10240000 correct (74.01)
Epoch 41, Iteration 0, loss = 0.0246
Epoch 41, Iteration 100, loss = 0.0343
Epoch 41, Iteration 200, loss = 0.1541
Epoch 41, Iteration 300, loss = 0.0710
Epoch 41, Iteration 400, loss = 0.0254
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Epoch 41, Iteration 500, loss = 0.0304
Epoch 41, Iteration 600, loss = 0.0257
Epoch 41, Iteration 700, loss = 0.0731
Eval 7471746 / 10240000 correct (72.97)
Epoch 42, Iteration 0, loss = 0.0186
Epoch 42, Iteration 100, loss = 0.0326
Epoch 42, Iteration 200, loss = 0.0405
Epoch 42, Iteration 300, loss = 0.0430
Epoch 42, Iteration 400, loss = 0.0204
Epoch 42, Iteration 500, loss = 0.0808
Epoch 42, Iteration 600, loss = 0.0401
Epoch 42, Iteration 700, loss = 0.0331
Eval 7440827 / 10240000 correct (72.66)
Epoch 43, Iteration 0, loss = 0.0264
Epoch 43, Iteration 100, loss = 0.0586
Epoch 43, Iteration 200, loss = 0.0446
Epoch 43, Iteration 300, loss = 0.0455
Epoch 43, Iteration 400, loss = 0.0250
Epoch 43, Iteration 500, loss = 0.0584
Epoch 43, Iteration 600, loss = 0.1510
Epoch 43, Iteration 700, loss = 0.0969
Eval 7595306 / 10240000 correct (74.17)
Epoch 44, Iteration 0, loss = 0.0163
Epoch 44, Iteration 100, loss = 0.0541
Epoch 44, Iteration 200, loss = 0.0294
Epoch 44, Iteration 300, loss = 0.0340
Epoch 44, Iteration 400, loss = 0.0314
Epoch 44, Iteration 500, loss = 0.0192
Epoch 44, Iteration 600, loss = 0.0299
Epoch 44, Iteration 700, loss = 0.1090
Eval 7605962 / 10240000 correct (74.28)
Epoch 45, Iteration 0, loss = 0.0182
Epoch 45, Iteration 100, loss = 0.0209
Epoch 45, Iteration 200, loss = 0.0167
Epoch 45, Iteration 300, loss = 0.1070
Epoch 45, Iteration 400, loss = 0.0342
Epoch 45, Iteration 500, loss = 0.0734
Epoch 45, Iteration 600, loss = 0.0229
Epoch 45, Iteration 700, loss = 0.0267
Eval 7499295 / 10240000 correct (73.24)
Epoch 46, Iteration 0, loss = 0.0201
Epoch 46, Iteration 100, loss = 0.0652
Epoch 46, Iteration 200, loss = 0.0445
Epoch 46, Iteration 300, loss = 0.0291
Epoch 46, Iteration 400, loss = 0.0428
Epoch 46, Iteration 500, loss = 0.1065
Epoch 46, Iteration 600, loss = 0.0567
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Epoch 46, Iteration 700, loss = 0.0302
Eval 7654877 / 10240000 correct (74.75)
Epoch 47, Iteration 0, loss = 0.0253
Epoch 47, Iteration 100, loss = 0.0843
Epoch 47, Iteration 200, loss = 0.0356
Epoch 47, Iteration 300, loss = 0.0421
Epoch 47, Iteration 400, loss = 0.0493
Epoch 47, Iteration 500, loss = 0.0948
Epoch 47, Iteration 600, loss = 0.0459
Epoch 47, Iteration 700, loss = 0.1196
Eval 7592488 / 10240000 correct (74.15)
Epoch 48, Iteration 0, loss = 0.0784
Epoch 48, Iteration 100, loss = 0.0487
Epoch 48, Iteration 200, loss = 0.0229
Epoch 48, Iteration 300, loss = 0.0478
Epoch 48, Iteration 400, loss = 0.0303
Epoch 48, Iteration 500, loss = 0.0540
Epoch 48, Iteration 600, loss = 0.0206
Epoch 48, Iteration 700, loss = 0.0342
Eval 7555538 / 10240000 correct (73.78)
Epoch 49, Iteration 0, loss = 0.0579
Epoch 49, Iteration 100, loss = 0.0217
Epoch 49, Iteration 200, loss = 0.0348
Epoch 49, Iteration 300, loss = 0.0942
Epoch 49, Iteration 400, loss = 0.0910
Epoch 49, Iteration 500, loss = 0.0434
Epoch 49, Iteration 600, loss = 0.0279
Epoch 49, Iteration 700, loss = 0.0251
Eval 7356789 / 10240000 correct (71.84)
Epoch 50, Iteration 0, loss = 0.0395
Epoch 50, Iteration 100, loss = 0.0427
Epoch 50, Iteration 200, loss = 0.0282
Epoch 50, Iteration 300, loss = 0.0423
Epoch 50, Iteration 400, loss = 0.0575
Epoch 50, Iteration 500, loss = 0.0221
Epoch 50, Iteration 600, loss = 0.0362
Epoch 50, Iteration 700, loss = 0.0398
Eval 7554613 / 10240000 correct (73.78)
Epoch 51, Iteration 0, loss = 0.0298
Epoch 51, Iteration 100, loss = 0.0486
Epoch 51, Iteration 200, loss = 0.0404
Epoch 51, Iteration 300, loss = 0.0164
Epoch 51, Iteration 400, loss = 0.0229
Epoch 51, Iteration 500, loss = 0.0205
Epoch 51, Iteration 600, loss = 0.0830
Epoch 51, Iteration 700, loss = 0.0402
Eval 7580798 / 10240000 correct (74.03)
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Epoch 52, Iteration 0, loss = 0.1160
Epoch 52, Iteration 100, loss = 0.0170
Epoch 52, Iteration 200, loss = 0.0258
Epoch 52, Iteration 300, loss = 0.0259
Epoch 52, Iteration 400, loss = 0.0240
Epoch 52, Iteration 500, loss = 0.0268
Epoch 52, Iteration 600, loss = 0.0328
Epoch 52, Iteration 700, loss = 0.0175
Eval 7526829 / 10240000 correct (73.50)
Epoch 53, Iteration 0, loss = 0.0224
Epoch 53, Iteration 100, loss = 0.0422
Epoch 53, Iteration 200, loss = 0.0456
Epoch 53, Iteration 300, loss = 0.0560
Epoch 53, Iteration 400, loss = 0.0450
Epoch 53, Iteration 500, loss = 0.0148
Epoch 53, Iteration 600, loss = 0.0768
Epoch 53, Iteration 700, loss = 0.0603
Eval 7341276 / 10240000 correct (71.69)
Epoch 54, Iteration 0, loss = 0.0106
Epoch 54, Iteration 100, loss = 0.1026
Epoch 54, Iteration 200, loss = 0.0698
Epoch 54, Iteration 300, loss = 0.0265
Epoch 54, Iteration 400, loss = 0.0211
Epoch 54, Iteration 500, loss = 0.0457
Epoch 54, Iteration 600, loss = 0.0186
Epoch 54, Iteration 700, loss = 0.0154
Eval 7601171 / 10240000 correct (74.23)
Epoch 55, Iteration 0, loss = 0.0225
Epoch 55, Iteration 100, loss = 0.0242
Epoch 55, Iteration 200, loss = 0.0489
Epoch 55, Iteration 300, loss = 0.0407
Epoch 55, Iteration 400, loss = 0.0353
Epoch 55, Iteration 500, loss = 0.0444
Epoch 55, Iteration 600, loss = 0.0375
Epoch 55, Iteration 700, loss = 0.1180
Eval 7635738 / 10240000 correct (74.57)
Epoch 56, Iteration 0, loss = 0.0195
Epoch 56, Iteration 100, loss = 0.0456
Epoch 56, Iteration 200, loss = 0.0643
Epoch 56, Iteration 300, loss = 0.0382
Epoch 56, Iteration 400, loss = 0.1433
Epoch 56, Iteration 500, loss = 0.0216
Epoch 56, Iteration 600, loss = 0.0198
Epoch 56, Iteration 700, loss = 0.0745
Eval 7527567 / 10240000 correct (73.51)
Epoch 57, Iteration 0, loss = 0.0230
Epoch 57, Iteration 100, loss = 0.0228
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Epoch 57, Iteration 200, loss = 0.0351
Epoch 57, Iteration 300, loss = 0.0338
Epoch 57, Iteration 400, loss = 0.0232
Epoch 57, Iteration 500, loss = 0.0437
Epoch 57, Iteration 600, loss = 0.0319
Epoch 57, Iteration 700, loss = 0.0676
Eval 7637865 / 10240000 correct (74.59)
Epoch 58, Iteration 0, loss = 0.0259
Epoch 58, Iteration 100, loss = 0.0131
Epoch 58, Iteration 200, loss = 0.0560
Epoch 58, Iteration 300, loss = 0.1640
Epoch 58, Iteration 400, loss = 0.0165
Epoch 58, Iteration 500, loss = 0.1050
Epoch 58, Iteration 600, loss = 0.0735
Epoch 58, Iteration 700, loss = 0.0524
Eval 7604063 / 10240000 correct (74.26)
Epoch 59, Iteration 0, loss = 0.0333
Epoch 59, Iteration 100, loss = 0.0179
Epoch 59, Iteration 200, loss = 0.0118
Epoch 59, Iteration 300, loss = 0.0194
Epoch 59, Iteration 400, loss = 0.0150
Epoch 59, Iteration 500, loss = 0.0924
Epoch 59, Iteration 600, loss = 0.0423
Epoch 59, Iteration 700, loss = 0.0970
Eval 7401042 / 10240000 correct (72.28)
Epoch 60, Iteration 0, loss = 0.0297
Epoch 60, Iteration 100, loss = 0.0968
Epoch 60, Iteration 200, loss = 0.0199
Epoch 60, Iteration 300, loss = 0.0365
Epoch 60, Iteration 400, loss = 0.0212
Epoch 60, Iteration 500, loss = 0.1201
Epoch 60, Iteration 600, loss = 0.0785
Epoch 60, Iteration 700, loss = 0.0282
Eval 7594061 / 10240000 correct (74.16)
Epoch 61, Iteration 0, loss = 0.0317
Epoch 61, Iteration 100, loss = 0.0222
Epoch 61, Iteration 200, loss = 0.0436
Epoch 61, Iteration 300, loss = 0.0874
Epoch 61, Iteration 400, loss = 0.0262
Epoch 61, Iteration 500, loss = 0.0270
Epoch 61, Iteration 600, loss = 0.0164
Epoch 61, Iteration 700, loss = 0.0339
Eval 7468443 / 10240000 correct (72.93)
Epoch 62, Iteration 0, loss = 0.0317
Epoch 62, Iteration 100, loss = 0.0466
Epoch 62, Iteration 200, loss = 0.0307
Epoch 62, Iteration 300, loss = 0.0192
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Epoch 62, Iteration 400, loss = 0.0201
Epoch 62, Iteration 500, loss = 0.0423
Epoch 62, Iteration 600, loss = 0.0192
Epoch 62, Iteration 700, loss = 0.0531
Eval 7573936 / 10240000 correct (73.96)
Epoch 63, Iteration 0, loss = 0.0183
Epoch 63, Iteration 100, loss = 0.0184
Epoch 63, Iteration 200, loss = 0.0267
Epoch 63, Iteration 300, loss = 0.0339
Epoch 63, Iteration 400, loss = 0.0144
Epoch 63, Iteration 500, loss = 0.0402
Epoch 63, Iteration 600, loss = 0.0662
Epoch 63, Iteration 700, loss = 0.0185
Eval 7509811 / 10240000 correct (73.34)
Epoch 64, Iteration 0, loss = 0.0139
Epoch 64, Iteration 100, loss = 0.0136
Epoch 64, Iteration 200, loss = 0.0288
Epoch 64, Iteration 300, loss = 0.0351
Epoch 64, Iteration 400, loss = 0.0379
Epoch 64, Iteration 500, loss = 0.0871
Epoch 64, Iteration 600, loss = 0.0241
Epoch 64, Iteration 700, loss = 0.0218
Eval 7367821 / 10240000 correct (71.95)
Epoch 65, Iteration 0, loss = 0.0235
Epoch 65, Iteration 100, loss = 0.0150
Epoch 65, Iteration 200, loss = 0.0272
Epoch 65, Iteration 300, loss = 0.0157
Epoch 65, Iteration 400, loss = 0.0364
Epoch 65, Iteration 500, loss = 0.1192
Epoch 65, Iteration 600, loss = 0.2213
Epoch 65, Iteration 700, loss = 0.0283
Eval 7504471 / 10240000 correct (73.29)
Epoch 66, Iteration 0, loss = 0.0257
Epoch 66, Iteration 100, loss = 0.0619
Epoch 66, Iteration 200, loss = 0.0147
Epoch 66, Iteration 300, loss = 0.0171
Epoch 66, Iteration 400, loss = 0.1062
Epoch 66, Iteration 500, loss = 0.0298
Epoch 66, Iteration 600, loss = 0.0603
Epoch 66, Iteration 700, loss = 0.0281
Eval 7631554 / 10240000 correct (74.53)
Epoch 67, Iteration 0, loss = 0.0132
Epoch 67, Iteration 100, loss = 0.0198
Epoch 67, Iteration 200, loss = 0.0297
Epoch 67, Iteration 300, loss = 0.0255
Epoch 67, Iteration 400, loss = 0.0565
Epoch 67, Iteration 500, loss = 0.0203
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Epoch 67, Iteration 600, loss = 0.0301
Epoch 67, Iteration 700, loss = 0.0194
Eval 7581536 / 10240000 correct (74.04)
Epoch 68, Iteration 0, loss = 0.0438
Epoch 68, Iteration 100, loss = 0.0162
Epoch 68, Iteration 200, loss = 0.0522
Epoch 68, Iteration 300, loss = 0.0158
Epoch 68, Iteration 400, loss = 0.0760
Epoch 68, Iteration 500, loss = 0.0948
Epoch 68, Iteration 600, loss = 0.1353
Epoch 68, Iteration 700, loss = 0.0399
Eval 7517181 / 10240000 correct (73.41)
Epoch 69, Iteration 0, loss = 0.0345
Epoch 69, Iteration 100, loss = 0.0285
Epoch 69, Iteration 200, loss = 0.0152
Epoch 69, Iteration 300, loss = 0.0351
Epoch 69, Iteration 400, loss = 0.0218
Epoch 69, Iteration 500, loss = 0.0295
Epoch 69, Iteration 600, loss = 0.0198
Epoch 69, Iteration 700, loss = 0.0575
Eval 7468852 / 10240000 correct (72.94)
Epoch 70, Iteration 0, loss = 0.0139
Epoch 70, Iteration 100, loss = 0.0190
Epoch 70, Iteration 200, loss = 0.0176
Epoch 70, Iteration 300, loss = 0.0199
Epoch 70, Iteration 400, loss = 0.0209
Epoch 70, Iteration 500, loss = 0.1389
Epoch 70, Iteration 600, loss = 0.0694
Epoch 70, Iteration 700, loss = 0.0110
Eval 7619890 / 10240000 correct (74.41)
Epoch 71, Iteration 0, loss = 0.0344
Epoch 71, Iteration 100, loss = 0.0117
Epoch 71, Iteration 200, loss = 0.0115
Epoch 71, Iteration 300, loss = 0.0898
Epoch 71, Iteration 400, loss = 0.0206
Epoch 71, Iteration 500, loss = 0.0408
Epoch 71, Iteration 600, loss = 0.0160
Epoch 71, Iteration 700, loss = 0.0358
Eval 7630100 / 10240000 correct (74.51)
Epoch 72, Iteration 0, loss = 0.0135
Epoch 72, Iteration 100, loss = 0.0238
Epoch 72, Iteration 200, loss = 0.1770
Epoch 72, Iteration 300, loss = 0.0233
Epoch 72, Iteration 400, loss = 0.0120
Epoch 72, Iteration 500, loss = 0.0161
Epoch 72, Iteration 600, loss = 0.0179
Epoch 72, Iteration 700, loss = 0.0226
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Eval 7654534 / 10240000 correct (74.75)
Epoch 73, Iteration 0, loss = 0.0114
Epoch 73, Iteration 100, loss = 0.0083
Epoch 73, Iteration 200, loss = 0.1296
Epoch 73, Iteration 300, loss = 0.0171
Epoch 73, Iteration 400, loss = 0.1236
Epoch 73, Iteration 500, loss = 0.0548
Epoch 73, Iteration 600, loss = 0.0206
Epoch 73, Iteration 700, loss = 0.0161
Eval 7591946 / 10240000 correct (74.14)
Epoch 74, Iteration 0, loss = 0.0120
Epoch 74, Iteration 100, loss = 0.0179
Epoch 74, Iteration 200, loss = 0.0445
Epoch 74, Iteration 300, loss = 0.0482
Epoch 74, Iteration 400, loss = 0.0126
Epoch 74, Iteration 500, loss = 0.0202
Epoch 74, Iteration 600, loss = 0.1120
Epoch 74, Iteration 700, loss = 0.0259
Eval 7665667 / 10240000 correct (74.86)
Epoch 75, Iteration 0, loss = 0.0171
Epoch 75, Iteration 100, loss = 0.0123
Epoch 75, Iteration 200, loss = 0.0119
Epoch 75, Iteration 300, loss = 0.1270
Epoch 75, Iteration 400, loss = 0.0467
Epoch 75, Iteration 500, loss = 0.0089
Epoch 75, Iteration 600, loss = 0.0347
Epoch 75, Iteration 700, loss = 0.0206
Eval 7616570 / 10240000 correct (74.38)
Epoch 76, Iteration 0, loss = 0.0288
Epoch 76, Iteration 100, loss = 0.1029
Epoch 76, Iteration 200, loss = 0.0264
Epoch 76, Iteration 300, loss = 0.0910
Epoch 76, Iteration 400, loss = 0.0213
Epoch 76, Iteration 500, loss = 0.0349
Epoch 76, Iteration 600, loss = 0.0733
Epoch 76, Iteration 700, loss = 0.0157
Eval 7603506 / 10240000 correct (74.25)
Epoch 77, Iteration 0, loss = 0.0254
Epoch 77, Iteration 100, loss = 0.0723
Epoch 77, Iteration 200, loss = 0.0444
Epoch 77, Iteration 300, loss = 0.0120
Epoch 77, Iteration 400, loss = 0.0428
Epoch 77, Iteration 500, loss = 0.0168
Epoch 77, Iteration 600, loss = 0.0142
Epoch 77, Iteration 700, loss = 0.0186
Eval 7600714 / 10240000 correct (74.23)
Epoch 78, Iteration 0, loss = 0.0253
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Epoch 78, Iteration 100, loss = 0.0131
Epoch 78, Iteration 200, loss = 0.0174
Epoch 78, Iteration 300, loss = 0.0561
Epoch 78, Iteration 400, loss = 0.0318
Epoch 78, Iteration 500, loss = 0.0147
Epoch 78, Iteration 600, loss = 0.0395
Epoch 78, Iteration 700, loss = 0.0236
Eval 7647931 / 10240000 correct (74.69)
Epoch 79, Iteration 0, loss = 0.0317
Epoch 79, Iteration 100, loss = 0.0128
Epoch 79, Iteration 200, loss = 0.0111
Epoch 79, Iteration 300, loss = 0.0444
Epoch 79, Iteration 400, loss = 0.0452
Epoch 79, Iteration 500, loss = 0.1117
Epoch 79, Iteration 600, loss = 0.1507
Epoch 79, Iteration 700, loss = 0.0272
Eval 7667607 / 10240000 correct (74.88)
Epoch 80, Iteration 0, loss = 0.0229
Epoch 80, Iteration 100, loss = 0.0146
Epoch 80, Iteration 200, loss = 0.0160
Epoch 80, Iteration 300, loss = 0.0152
Epoch 80, Iteration 400, loss = 0.0551
Epoch 80, Iteration 500, loss = 0.0260
Epoch 80, Iteration 600, loss = 0.0450
Epoch 80, Iteration 700, loss = 0.0120
Eval 7672467 / 10240000 correct (74.93)
Epoch 81, Iteration 0, loss = 0.0392
Epoch 81, Iteration 100, loss = 0.0376
Epoch 81, Iteration 200, loss = 0.0204
Epoch 81, Iteration 300, loss = 0.1097
Epoch 81, Iteration 400, loss = 0.0184
Epoch 81, Iteration 500, loss = 0.1525
Epoch 81, Iteration 600, loss = 0.0391
Epoch 81, Iteration 700, loss = 0.0220
Eval 7737080 / 10240000 correct (75.56)
Epoch 82, Iteration 0, loss = 0.0232
Epoch 82, Iteration 100, loss = 0.0571
Epoch 82, Iteration 200, loss = 0.0108
Epoch 82, Iteration 300, loss = 0.0378
Epoch 82, Iteration 400, loss = 0.0175
Epoch 82, Iteration 500, loss = 0.0191
Epoch 82, Iteration 600, loss = 0.0218
Epoch 82, Iteration 700, loss = 0.0158
Eval 7663417 / 10240000 correct (74.84)
Epoch 83, Iteration 0, loss = 0.0371
Epoch 83, Iteration 100, loss = 0.0318
Epoch 83, Iteration 200, loss = 0.0639
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Epoch 83, Iteration 300, loss = 0.0150
Epoch 83, Iteration 400, loss = 0.0171
Epoch 83, Iteration 500, loss = 0.0200
Epoch 83, Iteration 600, loss = 0.0289
Epoch 83, Iteration 700, loss = 0.0144
Eval 7680079 / 10240000 correct (75.00)
Epoch 84, Iteration 0, loss = 0.0150
Epoch 84, Iteration 100, loss = 0.0176
Epoch 84, Iteration 200, loss = 0.0434
Epoch 84, Iteration 300, loss = 0.0117
Epoch 84, Iteration 400, loss = 0.0237
Epoch 84, Iteration 500, loss = 0.0155
Epoch 84, Iteration 600, loss = 0.0312
Epoch 84, Iteration 700, loss = 0.0186
Eval 7630311 / 10240000 correct (74.51)
Epoch 85, Iteration 0, loss = 0.0268
Epoch 85, Iteration 100, loss = 0.0111
Epoch 85, Iteration 200, loss = 0.0198
Epoch 85, Iteration 300, loss = 0.0344
Epoch 85, Iteration 400, loss = 0.0130
Epoch 85, Iteration 500, loss = 0.0160
Epoch 85, Iteration 600, loss = 0.0672
Epoch 85, Iteration 700, loss = 0.0615
Eval 7526343 / 10240000 correct (73.50)
Epoch 86, Iteration 0, loss = 0.0312
Epoch 86, Iteration 100, loss = 0.0375
Epoch 86, Iteration 200, loss = 0.0743
Epoch 86, Iteration 300, loss = 0.0238
Epoch 86, Iteration 400, loss = 0.0323
Epoch 86, Iteration 500, loss = 0.0156
Epoch 86, Iteration 600, loss = 0.0405
Epoch 86, Iteration 700, loss = 0.0264
Eval 7640979 / 10240000 correct (74.62)
Epoch 87, Iteration 0, loss = 0.0186
Epoch 87, Iteration 100, loss = 0.0127
Epoch 87, Iteration 200, loss = 0.0387
Epoch 87, Iteration 300, loss = 0.0372
Epoch 87, Iteration 400, loss = 0.0151
Epoch 87, Iteration 500, loss = 0.0151
Epoch 87, Iteration 600, loss = 0.0149
Epoch 87, Iteration 700, loss = 0.0106
Eval 7559210 / 10240000 correct (73.82)
Epoch 88, Iteration 0, loss = 0.0406
Epoch 88, Iteration 100, loss = 0.0345
Epoch 88, Iteration 200, loss = 0.0173
Epoch 88, Iteration 300, loss = 0.0336
Epoch 88, Iteration 400, loss = 0.0094
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Epoch 88, Iteration 500, loss = 0.0144
Epoch 88, Iteration 600, loss = 0.0179
Epoch 88, Iteration 700, loss = 0.0152
Eval 7612711 / 10240000 correct (74.34)
Epoch 89, Iteration 0, loss = 0.0332
Epoch 89, Iteration 100, loss = 0.0130
Epoch 89, Iteration 200, loss = 0.0447
Epoch 89, Iteration 300, loss = 0.0255
Epoch 89, Iteration 400, loss = 0.0104
Epoch 89, Iteration 500, loss = 0.0145
Epoch 89, Iteration 600, loss = 0.0191
Epoch 89, Iteration 700, loss = 0.0785
Eval 7635068 / 10240000 correct (74.56)
Epoch 90, Iteration 0, loss = 0.0208
Epoch 90, Iteration 100, loss = 0.0127
Epoch 90, Iteration 200, loss = 0.0097
Epoch 90, Iteration 300, loss = 0.0202
Epoch 90, Iteration 400, loss = 0.0105
Epoch 90, Iteration 500, loss = 0.0557
Epoch 90, Iteration 600, loss = 0.0391
Epoch 90, Iteration 700, loss = 0.0127
Eval 7718079 / 10240000 correct (75.37)
Epoch 91, Iteration 0, loss = 0.0388
Epoch 91, Iteration 100, loss = 0.0263
Epoch 91, Iteration 200, loss = 0.0120
Epoch 91, Iteration 300, loss = 0.0193
Epoch 91, Iteration 400, loss = 0.0098
Epoch 91, Iteration 500, loss = 0.0292
Epoch 91, Iteration 600, loss = 0.0893
Epoch 91, Iteration 700, loss = 0.0187
Eval 7651322 / 10240000 correct (74.72)
Epoch 92, Iteration 0, loss = 0.0107
Epoch 92, Iteration 100, loss = 0.1110
Epoch 92, Iteration 200, loss = 0.0201
Epoch 92, Iteration 300, loss = 0.0076
Epoch 92, Iteration 400, loss = 0.0595
Epoch 92, Iteration 500, loss = 0.0673
Epoch 92, Iteration 600, loss = 0.0128
Epoch 92, Iteration 700, loss = 0.0562
Eval 7482369 / 10240000 correct (73.07)
Epoch 93, Iteration 0, loss = 0.0414
Epoch 93, Iteration 100, loss = 0.0475
Epoch 93, Iteration 200, loss = 0.0145
Epoch 93, Iteration 300, loss = 0.0131
Epoch 93, Iteration 400, loss = 0.0568
Epoch 93, Iteration 500, loss = 0.0218
Epoch 93, Iteration 600, loss = 0.0120
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Epoch 93, Iteration 700, loss = 0.0581
Eval 7596299 / 10240000 correct (74.18)
Epoch 94, Iteration 0, loss = 0.0145
Epoch 94, Iteration 100, loss = 0.0449
Epoch 94, Iteration 200, loss = 0.0140
Epoch 94, Iteration 300, loss = 0.0289
Epoch 94, Iteration 400, loss = 0.0210
Epoch 94, Iteration 500, loss = 0.1949
Epoch 94, Iteration 600, loss = 0.0341
Epoch 94, Iteration 700, loss = 0.0382
Eval 7610460 / 10240000 correct (74.32)
Epoch 95, Iteration 0, loss = 0.0107
Epoch 95, Iteration 100, loss = 0.0126
Epoch 95, Iteration 200, loss = 0.0140
Epoch 95, Iteration 300, loss = 0.0141
Epoch 95, Iteration 400, loss = 0.0604
Epoch 95, Iteration 500, loss = 0.0217
Epoch 95, Iteration 600, loss = 0.0147
Epoch 95, Iteration 700, loss = 0.0560
Eval 7648755 / 10240000 correct (74.69)
Epoch 96, Iteration 0, loss = 0.0116
Epoch 96, Iteration 100, loss = 0.0135
Epoch 96, Iteration 200, loss = 0.0134
Epoch 96, Iteration 300, loss = 0.0165
Epoch 96, Iteration 400, loss = 0.0538
Epoch 96, Iteration 500, loss = 0.0506
Epoch 96, Iteration 600, loss = 0.0169
Epoch 96, Iteration 700, loss = 0.0100
Eval 7617463 / 10240000 correct (74.39)
Epoch 97, Iteration 0, loss = 0.0092
Epoch 97, Iteration 100, loss = 0.0315
Epoch 97, Iteration 200, loss = 0.0472
Epoch 97, Iteration 300, loss = 0.0173
Epoch 97, Iteration 400, loss = 0.0152
Epoch 97, Iteration 500, loss = 0.0196
Epoch 97, Iteration 600, loss = 0.0498
Epoch 97, Iteration 700, loss = 0.0125
Eval 7620842 / 10240000 correct (74.42)
Epoch 98, Iteration 0, loss = 0.0978
Epoch 98, Iteration 100, loss = 0.0109
Epoch 98, Iteration 200, loss = 0.0908
Epoch 98, Iteration 300, loss = 0.0088
Epoch 98, Iteration 400, loss = 0.0196
Epoch 98, Iteration 500, loss = 0.0164
Epoch 98, Iteration 600, loss = 0.0160
Epoch 98, Iteration 700, loss = 0.0783
Eval 7571793 / 10240000 correct (73.94)
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Epoch 99, Iteration 0, loss = 0.0141
Epoch 99, Iteration 100, loss = 0.0170
Epoch 99, Iteration 200, loss = 0.0601
Epoch 99, Iteration 300, loss = 0.0227
Epoch 99, Iteration 400, loss = 0.0663
Epoch 99, Iteration 500, loss = 0.0465
Epoch 99, Iteration 600, loss = 0.0341
Epoch 99, Iteration 700, loss = 0.0879
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Eval 7657923 / 10240000 correct (74.78)

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Epoch 95, Iteration 300, loss = 0.0141
Epoch 95, Iteration 400, loss = 0.0604
Epoch 95, Iteration 500, loss = 0.0217
Epoch 95, Iteration 600, loss = 0.0147
Epoch 95, Iteration 700, loss = 0.0560
Eval 7648755 / 10240000 correct (74.69)
Epoch 96, Iteration 0, loss = 0.0116
Epoch 96, Iteration 100, loss = 0.0135
Epoch 96, Iteration 200, loss = 0.0134
Epoch 96, Iteration 300, loss = 0.0165
Epoch 96, Iteration 400, loss = 0.0538
Epoch 96, Iteration 500, loss = 0.0506
Epoch 96, Iteration 600, loss = 0.0169
Epoch 96, Iteration 700, loss = 0.0100
Eval 7617463 / 10240000 correct (74.39)
Epoch 97, Iteration 0, loss = 0.0092
Epoch 97, Iteration 100, loss = 0.0315
Epoch 97, Iteration 200, loss = 0.0472
Epoch 97, Iteration 300, loss = 0.0173
Epoch 97, Iteration 400, loss = 0.0152
Epoch 97, Iteration 500, loss = 0.0196
Epoch 97, Iteration 600, loss = 0.0498
Epoch 97, Iteration 700, loss = 0.0125
Eval 7620842 / 10240000 correct (74.42)
Epoch 98, Iteration 0, loss = 0.0978
Epoch 98, Iteration 100, loss = 0.0109
Epoch 98, Iteration 200, loss = 0.0908
Epoch 98, Iteration 300, loss = 0.0088
Epoch 98, Iteration 400, loss = 0.0196
Epoch 98, Iteration 500, loss = 0.0164
Epoch 98, Iteration 600, loss = 0.0160
Epoch 98, Iteration 700, loss = 0.0783
Eval 7571793 / 10240000 correct (73.94)
Epoch 99, Iteration 0, loss = 0.0141
Epoch 99, Iteration 100, loss = 0.0170
Epoch 99, Iteration 200, loss = 0.0601
Epoch 99, Iteration 300, loss = 0.0227
Epoch 99, Iteration 400, loss = 0.0663
Epoch 99, Iteration 500, loss = 0.0465
Epoch 99, Iteration 600, loss = 0.0341
Epoch 99, Iteration 700, loss = 0.0879
Eval 7657923 / 10240000 correct (74.78)
```